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Answers and Explanations for
MoE's Model Exit Exam
Questions administered on
July, 04 2023 for the
Department of Economics
(BA)



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Preface

Dear Exam Candidates,

This document contains the answers and explanations for 94 model the exit exam questions administered by the Ministry of Education on July 04, 2023 for the department of Economics. You can access more questions on our Telegram channel via this link: https://t.me/ethio_econ , or by scanning the QR code provided at the bottom of each page.

We want to disclam that EthioEcon is not affiliated to the Ministry of Education.

We wish you all the best in your preparation for the upcoming exit exam.

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EthioEcon



Question 1

Let y_0 be the amount of investment in period zero and MPC is the marginal propensity to consume. Income increment in period 1, i.e., $y_1 = MPCy_0$. Or If MPC is 0.75, $y_1 = 0.75y_0$. What is the magnitude of income increment in any time period t ?

- a. $y_t = 0.25^t y_0$
- b. $y_t = 0.75^t y_0$
- c. $y_t = 1.25^t y_0$
- d. $y_t = 0.75y_0$

Question Description	
Theme	Quantitative Methods in Economics with application
Course Name	Mathematical Economics
Chapter	Chapter 5: Difference Equations

Solution

In the simple multiplier analysis, a single investment expenditure in period 0 will call fourth successive rounds of spending, which in turn will bring about varying amounts of income increment in succeeding time periods. Using y to denote income increment, we have y_0 = the amount of investment in period 0; but the subsequent income increments will depend on the marginal propensity to consume (MPC). If $MPC = 0.75$ and if the income of each period is consumed only in the next period, then 75 percent of y_0 will be consumed in period 1, resulting in an income increment in period 1 of $y_1 = 0.75y_0$. By similar reasoning, we can find $y_2 = 0.75y_1$, etc. By iteration we have

$$y_1 = 0.75y_0$$

$$y_2 = 0.75y_1 = 0.75(0.75y_0) = (0.75)^2 y_0$$

$$y_3 = 0.75y_2 = 0.75(0.75)^2 y_0 = (0.75)^3 y_0$$

These can be summarized into the solution $y_t = (0.75)^t y_0$.

Hence, the correct answer is choice **(B)** $y_t = (0.75)^t y_0$.



Question 2

A country with a current account deficit is

- a. importing and exporting both current and future consumptions
- b. importing present consumption and exporting future consumption
- c. exporting present consumption and importing future consumption
- d. importing future consumption and exporting current consumption

Question Description	
Theme	Basic Macroeconomics
Course Name	International Economics II
Chapter	Chapter 3: Balance of Payments

Solution

When the autarky price is below relative price of future consumption, future consumption is relatively cheap in the home economy and present consumption is relatively expensive. Thus, home economy will “import” present consumption from abroad in the first period (by running a current account deficit) and “export” future consumption later (by repaying its foreign debt). This result is in accord with the principle of comparative advantage from international trade theory, which states that countries tend to import those commodities whose autarky prices are high compared with world prices and export those whose autarky prices are comparatively low.

Hence the correct answer is **(B)** importing present consumption and exporting future consumption.

Exam Tip

Note that choice (C) and (D) are the same except the word “current” is used in place of “present” in choice (D). This rules out both choices as an answer. Hence it is helpful to scrutinize questions in the exam even if you do not fully comprehend what the question is really asking.



Question 3

According to Keynesian economists

- a. The economy is inherently stable so that aggregate demand remain unaffected
- b. General equilibrium is attained relatively quickly because the price level adjusts quickly.
- c. The government should only monitor the economy and pump money in which it finds a slump is imminent.
- d. General equilibrium is not attained relatively quickly because the price level adjusts slowly.

Question Description	
Theme	Basic Macroeconomics
Course Name	Macroeconomics I
Chapter	Chapter 1: The state of Macroeconomics

Solution

Keynesian economics: the macroeconomic generalizations that lead to the conclusion that a capitalistic economy is characterized by macroeconomic instability and that fiscal policy and monetary policy can be used to promote full employment, price-level stability, and economic growth.

Keynesian economists believe that the self-correcting feature of the market, which is of course the hallmark of classical theory, does not work. They think that prices and especially wages respond slowly to changes in supply and demand, resulting in shortages and surpluses.

With regard to the role of the government, Keynes argued that the government role was needed to preserve capitalism because without management, a modern capitalist economy is so unstable that it may threaten the social compact that it rests on.

Hence the correct answer is choice **(D)** Keynesian Economists believe that general equilibrium is not attained relatively quickly because the price level adjusts slowly (is sticky).



Question 4

According to the structuralism model of economic growth,

- a. Given the structural similarities LDCs cannot replicate what DCs have done in the past. Thus, both LDCS and DCs can follow similar growth trajectory
- b. Given the structural differences LDCs replicate what DCs have done in the past. Thus, both LDCS and DCs can follow similar growth trajectory
- c. Given the structural similarities LDCs cannot replicate what DCs have done in the past. Thus, both LDCS and DCs cannot follow similar growth trajectory
- d. Given the structural differences LDCs cannot replicate what DCs have done in the past. Thus, both LDCS and DCs cannot follow similar growth trajectory

Question Description	
Theme	Basic Macroeconomics
Course Name	Development economics I
Chapter	Chapter 3: Growth Models and Theories of development

Solution

The essence of structuralism is that the LDCs should focus their development effort within themselves by mobilizing their internal potentials and resources. Structuralism tended to blame the increasing income disparities between the developed countries (DCs) and the Least developed countries (LDCs) on the legitimacy accorded to the existing pattern of world economic order by neo classical models and tenets.

Structuralist economic thinking is driven by the notions of the peculiarities of the particular economic systems and their structural circumstances. It is based on the notion that development must be fashioned in accord with the economy's particular situation: level of development at the time, historical circumstances, geographical conditions, political setting, and the prevailing and overall global economic environment.

Hence the correct answer is **(D)** Given the structural differences LDCs cannot replicate what DCs have done in the past. Thus, both LDCs and DCs cannot follow similar growth trajectory.



Question 5

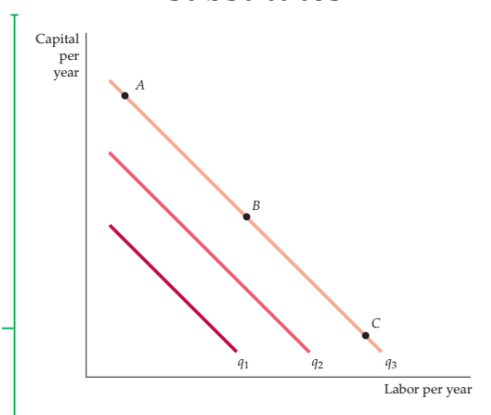
An L-shaped isoquants represents that

- a. Inputs must be used in variable proportions
- b. There is only one efficient production process
- c. All production processes are efficient
- d. Inputs are perfect substitutes

Question Description	
Theme	Basic Microeconomics
Course Name	Microeconomics I
Chapter	Chapter 3: The Theory of Production

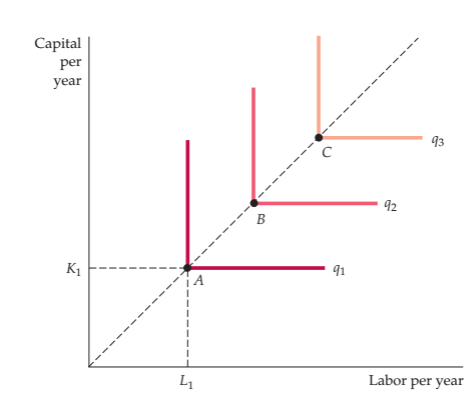
Solution

Isoquants when inputs are perfect substitutes



When the isoquants are straight lines, the MRTS is constant. Thus the rate at which capital and labor can be substituted for each other is the same no matter what level of inputs is being used. Points A, B, and C represent three different capital-labor combinations that generate the same output q_3 .

Fixed-proportions (Leontief) production function



When the isoquants are L-shaped, only one combination of labor and capital can be used to produce a given output (as at point A on isoquant q_1 , point B on isoquant q_2 , and point C on isoquant q_3). Adding more labor alone does not increase output, nor does adding more capital alone.

Hence the correct answer is **(B)** there is only one efficient production process.



Question 6

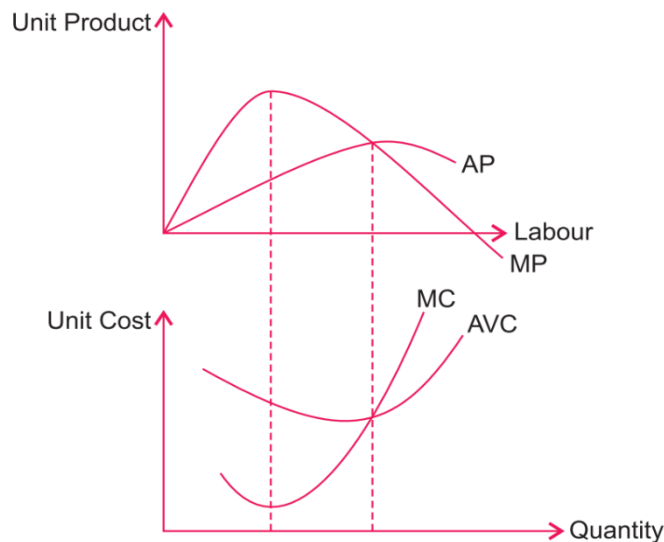
At a point where average variable cost is minimum, which of the following is correct?

- a. Marginal product attains its maximum
- b. Total product is maximized
- c. The law of diminishing returns begins to operate
- d. Average product is at its maximum

Question Description	
Theme	Basic Microeconomics
Course Name	Microeconomics I
Chapter	Chapter 4: Theory of Costs

Solution

The relationship between cost and production curves can be summarized by the diagram below.



The following important points can be taken away from the graph.

- TVC and TP of labor are inversely related.
- AVC and AP of labor are inversely related.
- MC and MP of labor are inversely related.

Hence the answer is **(D)** Average product is at its maximum.



Question 7

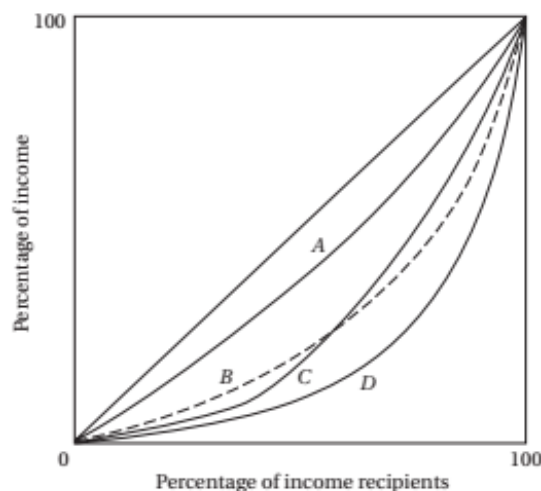
Based on the analyses of the described Lorenz curves above, which one of the following can be concluded about the two countries?

- For every percentage of income the percentage of income recipients in Country B are greater than those in country A
- For every percentage of income the percentage of income recipients in Country A are greater than those in country B
- For every percentage of income the percentage of income recipients in Country A and in country B cannot be known from the above information.
- For every percentage of income the percentage of income recipients in Country A are equal to those in country B

Question Description	
Theme	Basic Macroeconomics
Course Name	Development economics I
Chapter	Chapter 5: Income Inequality, Poverty and Development

Solution

Even though the Lorenz curves were not visible during the exam, let us use the Lorenz curves below to study the general properties. In the “Lorenz criterion”



of income distribution, whenever one Lorenz curve lies above another Lorenz curve, the economy corresponding to the upper Lorenz curve is more equal than that of the lower curve. Thus, economy A may unambiguously be said to be more equal than economy D. Whenever two Lorenz curves cross, such as curves B and C, the Lorenz criterion states that we “need more information” or additional assumptions before we can determine

which of the underlying economies is more equal.

Hence, had the question included a similar Lorenz curve, the answer would have been (A) For every percentage of income the percentage of income recipients in country B are greater than those in country A.



Question 8

Choose one of the key differences between the Consumer price index and The GDP deflator as a measure of the overall level of prices

- a. The CPI includes only those goods produced domestically.
- b. The CPI measures the prices of all goods and services produced, whereas the GDP deflator measures the prices of only the goods and services bought by consumers.
- c. The GDP deflator assigns fixed weights to the prices of different goods, whereas the CPI assigns changing weights
- d. GDP deflator allows the basket of goods to change over time as the composition of GDP changes

Question Description	
Theme	Basic Macroeconomics
Course Name	Macroeconomics I
Chapter	Chapter 2: National Income Accounting

Solution

The GDP deflator and the CPI give somewhat different information about what's happening to the overall level of prices in the economy.

- The first difference is that the GDP deflator measures the prices of all goods and services produced, whereas the CPI measures the prices of only the goods and services bought by consumers. Thus, an increase in the price of goods bought only by firms or the government will show up in the GDP deflator but not in the CPI.
- The second difference is that the GDP deflator includes only those goods produced domestically. Imported goods are not part of GDP and do not show up in the GDP deflator.
- The third and most subtle difference results from the way the two measures aggregate the many prices in the economy. The CPI assigns fixed weights to the prices of different goods, whereas the GDP deflator assigns changing weights. In other words, the CPI is computed using a fixed basket of goods, whereas the GDP deflator allows the basket of goods to change over time as the composition of GDP changes.

Hence the answer is **(D)** GDP deflator allows the basket of goods to change over time as the composition of GDP changes.



Question 9

Consider the coefficient for price elasticity of demand, $e_d = 5$ and the profit maximizing price set by the monopolist, $p = 15$. What would be its marginal cost at the optimum?

- a. 10
- b. 12
- c. 25
- d. 20

Question Description	
Theme	Basic Microeconomics
Course Name	Microeconomics I
Chapter	Chapter 6: Pure Monopoly Market

Solution

- The general expression for Marginal Revenue

$$MR = \frac{\Delta R}{\Delta Q} = \frac{\Delta(PQ)}{\Delta Q}$$

- Applying the product rule of derivative

$$MR = P + Q \frac{\Delta P}{\Delta Q} = P + P \left(\frac{Q}{P} \right) \left(\frac{\Delta P}{\Delta Q} \right)$$

- Substituting elasticity
- Equating MR and MC

$$MR = P + P(1/E_d)$$

$$P + P(1/E_d) = MC$$

$$MC = P + P \left(\frac{1}{e_d} \right)$$

$$MC = 15 + 15 \left(-\frac{1}{5} \right)$$

$$MC = 15 - 3$$

$$\underline{\underline{MC = 12}}$$

Note that the price elasticity of demand is negative and -5 should be used instead of 5 when substituting elasticity. In fact, you might find the above formula written as $MC = P - P \left(\frac{1}{e_d} \right)$ in some books which already consider the negative slope of demand curve at the beginning of the derivation process.

Hence the answer is **(B) 12**.



Question 10

Consider the following Single Linear Regression Model of Keynesian consumption function where household's consumption ($Cons_i$) is determined by income (INC_i) · β_0 is the intercept and the slope parameters respectively, ε_i is the disturbance term, i denotes the household i .

The income consumption data is collected from 100 households and fitted to the following econometric model

$$Cons_i = \beta_0 + \beta_1 INC_i + \varepsilon_i$$

The model is estimated and the result is obtained as follows and all parameters are statistically significant.

$$\widehat{Cons}_i = -124.84 + 0.853 INC_i$$

$$R^2 = 0.692$$

What is the interpretation of the assumption that "the relationship between $Cons_i$ and INC_i is linear?"

- The model is linear in income and the slope parameter
- The model is linear in consumption and the intercept parameter
- The model is linear in consumption and income
- The model is linear in the intercept and the slope parameter

Question Description	
Theme	Quantitative Methods in Economics with application
Course Name	Econometrics I
Chapter	Chapter 2: Simple Linear Regression

Solution

One of the assumptions underlying the method of least squares in The classical linear regression models is linearity in parameters, though it may or may not be linear in the variables. Linear in parameter means the β 's (that is the parameters) are raised to the first power only.

Hence the answer is **(D)** The model is linear in the intercept and slope parameter.



Question 11

Considering the same estimated utility function $U = q_1^{0.4}q_2^{0.6}$ for the consumer and the prices of q_1 and q_2 are $P_1 = 6$ and $P_2 = 10$, respectively, and she spends birr 144 on these goods, what are the optimal numbers of q_1 and q_2 that she will buy?

- a. $q_1 = 10.6$ and $q_2 = 8$
- b. $q_1 = 12$ and $q_2 = 7.2$
- c. $q_1 = 6$ and $q_2 = 8$
- d. $q_1 = 10$ and $q_2 = 8.4$

Question Description	
Theme	Basic Microeconomics
Course Name	Microeconomics I
Chapter	Chapter 1: Theory of Consumer Behavior and Demand

Solution

$$MU_1 = \frac{\partial U}{\partial q_1} = 0.4q_1^{-0.6}q_2^{0.6} \text{ (Equation 1)}$$

$$MU_2 = \frac{\partial U}{\partial q_2} = 0.6q_1^{0.4}q_2^{-0.4} \text{ (Equation 2)}$$

The utility maximized in ordinal utility approach when $\frac{MU_1}{P_1} = \frac{MU_2}{P_2}$

$$\frac{0.4q_1^{-0.6}q_2^{0.6}}{6} = \frac{0.6q_1^{0.4}q_2^{-0.4}}{10}$$

Equating the ratio of Equation (1) and (2) with respect to their prices, we get

$$q_2 = 0.9q_1 \text{ (Equation 3)}$$

Based on the information in the question the budget line is

$$6q_1 + 10q_2 = 144 \text{ (Equation 4)}$$

Substituting Equation (3) in Equation (4) we get $q_1 = 9.6$ and $q_2 = 8.64$.

Hence, the answer to the question is **not provided** in the choices.



Question 12

Demand for foreign exchange is

- a. Past demand
- b. Derived demand
- c. Future demand
- d. Direct demand

Question Description	
Theme	Basic Macroeconomics
Course Name	International Economics II
Chapter	Chapter 1: The Foreign Exchange Market

Solution

The current account-based demand for foreign currency results from the desire to purchase goods and services from another country and to make unilateral transfers. In a sense, the demand for foreign currency is a secondary or derived demand because the foreign currency is a means to acquiring something else.

Hence, the answer is **(B)** Derived demand.



Question 13

Dispute settlement among members is one of the functions of WTO. The need to settle disputes arises from _____.

- a. Political or other disagreements among members
- b. Unfulfilled promises with respect to one or more of the WTO agreements
- c. Economic policies that extend beyond trade
- d. Consumer and producer preference on the type of goods traded

Question Description	
Theme	Basic Macroeconomics
Course Name	International Economics I
Chapter	Chapter 5: International Trade Institutions and Regional Economic Arrangements

Solution

The most important new aspect of the WTO is generally acknowledged to be its “dispute settlement” procedure. A basic problem arises when one country accuses another of violating the rules of the trading system.

A dispute can arise when one country adopts a trade policy or violates the WTO agreements. The WTO member countries use a multilateral system of settling disputes instead of taking actions unilaterally, meaning abiding by the agreed WTO procedures and respecting judgments. However, the WTO's priority is to settle disputes through consultations and not to pass a judgment.

Hence the answer is **(B)** unfulfilled promises with respect to one or more of the WTO agreements.



Question 14

Econometrics estimation of a model that emerges from the normal Cumulative Distribution Function(CDF) of the error term is known as with dummy dependent variable is understood s

- a. Probit Model
- b. Tobit Model
- c. Linear probability model
- d. Logit Model

Question Description	
Theme	Quantitative Methods in Economics with application
Course Name	Econometrics II
Chapter	Chapter 1: Regression Analysis with Qualitative Data

Solution

There are four approaches to developing a probability model for a binary response variable:

1. The linear probability model (LPM): is the simplest possible binary regression model in which the binary response variable is regressed on the relevant explanatory variables by using the standard OLS methodology.
2. The logit model: in this model the dependent variable is the log of the odds ratio, which is a linear function of the regressors. The probability function that underlies the logit model is the logistic distribution.
3. The probit (normit) model: emerges from normal cumulative distribution functions (CDF). This model is mathematically a bit difficult as it involves integrals.
4. The tobit model: is closely related to the probit model and is also known as a censored regression model. In this model, the response variable is observed only if a certain condition(s) is met.

Hence the answer is **(A)** Probit Model.



Question 15

Fiat money is

- a. Backed by government
- b. Backed by gold
- c. Real import
- d. Backed by real export

Question Description	
Theme	Basic Macroeconomics
Course Name	Monetary Economics
Chapter	Chapter 1: Money and Monetary Theories

Solution

Fiat money, in a broad sense, all kinds of money that are made legal tender by a government decree or fiat. The term is, however, usually reserved for legal-tender paper money or coins that have face values far exceeding their commodity values and are not redeemable in gold or silver.

The word “fiat” is Latin in origin and refers to an arbitrary order issued by a government or other authoritative figure. When applied to paper money, fiat currency refers to the scary notion that our dollar has value only because the government says it does.

Hence the answer is **(A)** Backed by government.



Question 16

Firms' demand for money arising precautionary motive is due to one of the following:

- a. Predicted demand increases for their products
- b. Predicted market conditions
- c. sudden technological changes
- d. Pre-announced tax policies

Question Description	
Theme	Basic Macroeconomics
Course Name	Monetary Economics
Chapter	Chapter 3: The demand for money and other assets

Solution

According to Keynes's Liquidity Preference Theory, there are three motives behind the demand for money.

1. Transactions Motive: individuals are assumed to hold money because it is a medium of exchange that can be used to carry out everyday transactions.
2. Precautionary Motive: people hold money as a cushion against an unexpected need.
3. Speculative Motive: people hold money as a store of wealth.

Hence the answer is (C) sudden technological changes.

Additional Note

Keynes's resulting liquidity preference theory views that the transactions and precautionary components of money demand are proportional to income. However, the speculative component of money demand is viewed as sensitive to interest rates as well as to the expectations about the future movements of interest rates. This theory, then, implies that velocity is unstable and cannot be treated as a constant.



Question 17

Given a homogenous production function $Y = AK^{0.9}L^{0.5}$ compute the returns to scale if every factor is increased by β times and state whether increasing, decreasing or constant! When Factors increase by β times, Y increases by

- $\beta^{0.4}$ times and therefore the returns to scale is decreasing
- $\beta^{0.4}$ times and therefore the returns to scale is almost constant
- $\beta^{1.4}$ times and therefore the returns to scale is increasing
- $\beta^{0.45}$ times and therefore the returns to scale is decreasing

Question Description	
Theme	Basic Microeconomics
Course Name	Microeconomics I
Chapter	Chapter 3: The Theory of Production

Solution

- If raising all inputs (independent variables) k-fold raises the output by more than k-fold, it is an increasing returns to scale.
- If raising all inputs (independent variables) k-fold raises the output exactly k-fold also, it is a constant returns to scale.
- If raising all inputs (independent variables) k-fold raises the output by less than k-fold, it is a decreasing returns to scale.

For the given homogeneous production function $Y = AK^{0.9}L^{0.5}$, if all inputs are increased β times,

$$Y(\beta K, \beta L) = A(\beta K)^{0.9}(\beta L)^{0.5}$$

$$Y(\beta K, \beta L) = \beta^{0.9+0.5}AK^{0.9}L^{0.5}$$

$$Y(\beta K, \beta L) = \beta^{1.4}AK^{0.9}L^{0.5}$$

Hence, the answer is **(C)** $\beta^{1.4}$ times and therefore the returns to scale is increasing.



Question 18

Given the following information of a multiple regression model:

The following is demand model with 2 explanatory variables X_2 and X_3 , estimated over 15 observations using OLS where

y_i = quantity demanded of commodity i

X_2 = own price of commodity 2

X_3 = income of the individual i

$$y_i = \beta_1 + \beta_2 X_2 + \beta_3 X_3 + \varepsilon_i$$

And the following data have been calculated from the original X s

$$(X^T X)^{-1} = \begin{bmatrix} 2 & 3.5 & -1 \\ 3.5 & 1 & 6.5 \\ -1 & 6.5 & 4 \end{bmatrix} \quad \text{And } (X^T Y) = \begin{bmatrix} -3 \\ 2.2 \\ 0.6 \end{bmatrix}$$

Regression variance is obtained as 0.91

Compute the estimated values of the intercept and the slope coefficients from

$$\begin{bmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \\ \hat{\beta}_3 \end{bmatrix} = (X^T X)^{-1} (X^T Y)$$

What is the variance of $\hat{\beta}_3$

- a. -0.91
- b. 1.82
- c. 3.91
- d. 0.91

Question Description	
Theme	Quantitative Methods in Economics with application
Course Name	Econometrics I
Chapter	Chapter 3: Multiple Linear Regression



Solution

When elements in the principal diagonal of $(X^T X)^{-1}$ are multiplied by the regression variance, they give the regression parameters.

$$(X^T X)^{-1} = \begin{bmatrix} 2 & 3.5 & -1 \\ 3.5 & 1 & 6.5 \\ -1 & 6.5 & 4 \end{bmatrix}$$

$$\text{Var}(\hat{\beta}_1) = 0.91 \times 2 = 1.82$$

$$\text{Var}(\hat{\beta}_2) = 0.91 \times 1 = 0.91$$

$$\text{Var}(\hat{\beta}_3) = 0.91 \times 4 = 3.64$$

Hence, although choice (C) is the closest to the actual value, the exact answer to the question is **not provided** in the choices.

Additional information

The parameters could also be estimated from the matrix by multiplication.

$$\beta = (X^T X)^{-1} (X^T Y) = \begin{bmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \\ \hat{\beta}_3 \end{bmatrix} = \begin{bmatrix} 2 & 3.5 & -1 \\ 3.5 & 1 & 6.5 \\ -1 & 6.5 & 4 \end{bmatrix} \begin{bmatrix} -3 \\ 2.2 \\ 0.6 \end{bmatrix} = \begin{bmatrix} 1.1 \\ -4.4 \\ 19.7 \end{bmatrix}$$

Hence the demand model can be written as $y_t = 1.1 - 4.4X_2 + 19.7X_3$

Interpretation: A 1 unit increase in own price of the commodity, decreases the quantity demanded of the commodity by 4.4 units on average holding the income of the individual constant.

A 1 unit increase in the income of individual by 1 unit, increases the quantity demanded of the commodity by 19.7 units on average holding own price of the commodity constant.

Exam Tip

Do not be scared off by seemingly complicated questions. As you saw in the question above, some questions might only require a simple multiplication to arrive at the answer even though too much information is given to distract candidates.



Question 19

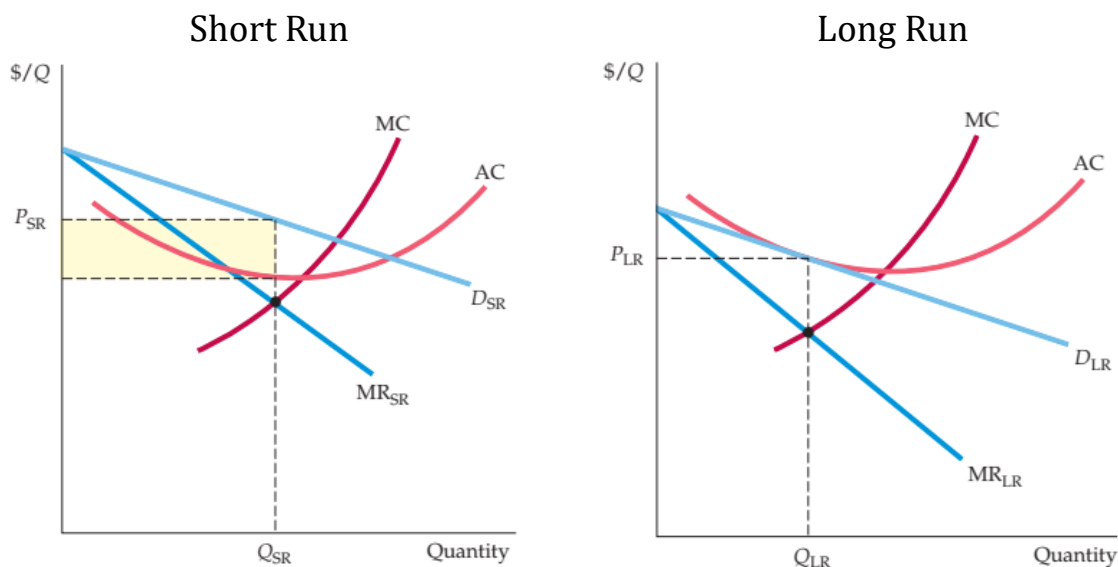
Given the long run perceived demand curve $LAC(q) = q^2 - 15q + 50$ and its long run average total cost curve for a firm in a monopolistic competition, what is the long run equilibrium price and output, respectively?

- a. 12 and 6
- b. 7 and 37
- c. 37 and 12
- d. 6 and 12

Question Description	
Theme	Basic Microeconomics
Course Name	Microeconomics II
Chapter	Chapter 1: Monopolistic Competition

Solution

Although the demand curve is missing in the exam question, let us try to study how a similar question would have been approached. Shown below are the curves for a monopolistically competitive firm in the short and long run.



As it can be seen in the curves above, monopolistically competitive firms earn zero profit (normal profit) in the long run. Hence all we need to do is equate the LAC and D_{LR} to determine the long run equilibrium price and output.



Suppose the long run derived demand curve is $P = 44 - 4q$ and the long run average total cost curve is $LAC(q) = q^2 - 20q + 108$ for a firm in monopolistic competition, the long run equilibrium price and output can be calculated as follows.

In the long run,

$$LAC = P$$

$$q^2 - 20q + 108 = 44 - 4q$$

$$q^2 - 16q + 64 = 0$$

$$(q - 8)^2 = 0$$

$$q = 8$$

Substituting the quantity in the demand curve yields

$$P = 44 - 4(8) = 12$$

Hence, the long run equilibrium price and output respectively is 12 and 8.

Additional Note

- Monopolistic competition is a market structure in which large number of firms compete by selling differentiated products.
- The demand curve of a firm under monopolistic competition is flatter or more price-elastic than that of a monopolistic firm.
- In the short run, monopolistic competition firms act as a monopolist and optimal decision by equating marginal revenue with marginal cost.
- Monopolistic competition firm in the long run make output decision at higher cost. It is produced at a point before the minimum long run average cost. As a result, firms in monopolistic competition industry said to be inefficient. They could not use the full capacity.



Question 20

Given: an initial investment of birr 60,000 and the after tax cash flows for year 1 of birr 8,000; for year 2 of birr 15,000; for year 3 of birr 22,000; for year 4 of birr 20,000 and for year 5 of birr 20,000: the payback period for this project is__

- a. 3.5 years
- b. 4 years
- c. 3.25 years
- d. 3.75 years

Question Description	
Theme	Basic Microeconomics
Course Name	Development Planning and Project Analysis II
Chapter	Chapter 2: Financial Analysis and Appraisal of Projects

Solution

Time	Cash Flow	Cumulative Sum
Year 1	Birr 8,000	
Year 2	Birr 15,000	Birr 23,000
Year 3	Birr 22,000	Birr 45,000
Year 4	Birr 20,000	Birr 65,000
Year 5	Birr 20,000	Birr 85,000

At the end of the 3rd year birr 45,000 out of the 60,000 initial investment has been returned. The remaining 15,000 birr will be returned in the fourth year where the cash flow is 20,000. Therefore, it takes $\frac{15,000}{20,000} = 0.75$ year (9 months) to completely pay back the remaining amount.

Hence the answer is **(D)** 3.75 years.



Question 21

Harris-Todaro migration model attempts to explain the process of achieving unemployment equilibrium between urban expected wages and average rural income rather than an equalized rural-urban wage. Which of the following is not a necessary assumption?

- a. Urban wages are institutionally determined and inflexible downward ,
- b. The decision to migrate also depends on political and social factors that cannot be quantified
- c. Rural workers are free to migrate and are willing to take their chances in the urban job lottery
- d. Migration is stimulated primarily by rational economic considerations of relative benefits and costs—mostly financial but also psychological.

Question Description	
Theme	Basic Macroeconomics
Course Name	Development economics II
Chapter	Chapter 3: Rural-urban interaction, migration and unemployment

Solution

According to the Harris-Todaro Migration model

- Migration is stimulated primarily by rational economic considerations of relative benefits and costs, mostly financial but also psychological.
- The decision to migrate depends on expected rather than actual urban rural real-wage differentials where the expected differential is determined.
- The probability of obtaining an urban job is directly related to the urban employment rate and thus inversely related to the urban unemployment rate.
- Urban wages are institutionally determined (inflexible downward) as assumed by Todaro.
- If rural workers were free to migrate, then despite the availability of only a few jobs, they are willing to take their chances in the urban job lottery.

Hence the answer is **(B)** The decision to migrate also depends on political and social factors that cannot be quantified.



Question 22

If an economic variable of interest Y at time t and time $t+1$ can be described using two parameters u and v as $uy_{t+1} = vy_t$ which one of the following expression correctly relate the time path of y_t ?

- a. $y_t = \left(\frac{u}{v}\right)^t y_0$
- b. $y_t = \frac{v}{u} y_0$
- c. $y_t = \left(\frac{v}{u}\right)^t y_0$
- d. $y_t = \left(\frac{v}{u}\right) y_0 t$

Question Description	
Theme	Quantitative Methods in Economics with application
Course Name	Mathematical Economics
Chapter	Chapter 5: Difference Equations

Solution

The homogeneous difference equation $uy_{t+1} = vy_t$ could be rewritten as

$$y_{t+1} = \left(\frac{v}{u}\right) y_t$$

which is the same as the equation in question 1 except for the replacement of 0.75 by v/u . Hence by analogy, the solution should be

$$y_t = \left(\frac{v}{u}\right)^t y_0$$

Hence the answer is (C) $y_t = \left(\frac{v}{u}\right)^t y_0$



Question 23

If consumers do not know how prices vary across firms

- a. Everyone benefits
- b. Their ignorance about price creates market power
- c. It may lead to a competitive price
- d. A firm can raise its prices by considerably losing its customers

Question Description	
Theme	Basic Microeconomics
Course Name	Microeconomics I
Chapter	Chapter 6: Pure Monopoly Market

Solution

One of the assumptions in a perfectly competitive market is “Perfect Knowledge”. It means that all sellers and buyers have complete knowledge of the conditions of the market. This knowledge refers not only to the prevailing conditions in the current period but in all future periods as well. But if consumers do not know how prices vary across firms, firms will gain market power.

Firms with market power are in an enviable position because they have the potential to earn large profits. Realizing that potential, however, may depend critically on pricing strategy. Even if the firm sets a single price, it needs an estimate of the elasticity of demand for its output. Ideally, the firm would like to price discriminate perfectly—i.e., to charge each customer his or her reservation price. In practice, this is almost always impossible. On the other hand, various forms of imperfect price discriminations are used to increase profits.

Hence the answer is **(B)** Their ignorance about price creates market power.



Question 24

If the government wants to attract investment, which of the following policy actions by the central bank can help the government to achieve its objective?

- a. Decrease interest rate
- b. Increase income tax
- c. Increase profit tax
- d. Increase interest rate

Question Description	
Theme	Basic Macroeconomics
Course Name	Monetary Economics
Chapter	Chapter 5: Central Banking and Monetary Policy

Solution

Major instruments of expansionary monetary policy, are:

- Bank Rate (Reduce it): Bank rate (also called discount rate) is the rate of interest at which the central bank lends to the commercial banks. Those banks in turn increase or decrease lending rates of interest accordingly. To check depression, the central bank reduces the bank rate, thereby enabling the commercial banks to take more loans from it and in turn to give more loans to producers at lower interest rates.
- Open Market Operation (Buy securities): These refer to the buying and selling of government securities which influence money supply in the economy. During depression, the central bank buys government bonds and securities from commercial banks, pay in cash to increase their cash stock and lending capacity.
- Cash-Reserve Ratio (Reduce CRR): Every commercial bank is required to keep with the central bank a particular percentage of its deposits or reserves in the form of cash. This percentage is called the cash reserve ratio (CRR). During depression, the central bank lowers the CRR, thereby increasing commercial banks' capacity to give credit.

Hence the answer is (A) Decrease interest rate.



Question 25

If the Hamiltonian function to be maximize is given as $H = 3y - u + \lambda(y + u)$, What is the optimal control path $\frac{\partial H}{\partial u}$, if $\lambda(2) = 0$

- a. $-1 - \lambda$, where $\lambda = 4e^{2-t} - 4$
- b. $\lambda - 1$, where $\lambda = 4e^{2-t} - 3$
- c. $\lambda - 2$, where $\lambda = 3e^{2-t} - 3$
- d. $\lambda - 1$, where $\lambda = 3e^{2-t} - 3$

Question Description	
Theme	Quantitative Methods in Economics with application
Course Name	Mathematical Economics
Chapter	Chapter 6: Dynamic Optimization

Solution

$$H = 3y - u + \lambda(y + u)$$

$$H = 3y - u + \lambda y + \lambda u$$

$$H = (3 + \lambda)y + (\lambda - 1)u$$

The Hamiltonian function above is linear in u . If we plot H against u in the uH plane, we get a straight line with slope $\frac{\partial H}{\partial u} = \lambda - 1$.

The costate equation is

$$\lambda' = -\frac{\partial H}{\partial y} = -3 - \lambda$$

The above equation can be rewritten as $\lambda' + \lambda = -3$

Note

For a nonhomogeneous linear differential equation $\frac{dy}{dt} + ay = b$, the general solution is given by $y(t) = Ae^{-at} + \frac{b}{a}$ ($a \neq 0$)



Based on the formula provided above, the general solution to $\lambda' + \lambda = -3$ is

$$\lambda(t) = Ae^{-t} - 3$$

Where A is an arbitrary constant. Using the transversality condition $\lambda(2) = 0$, we can calculate A.

$$0 = Ae^{-2} - 3$$

$$A = 3e^2$$

Thus, the definite solution for λ is

$$\lambda(t) = 3e^{2-t} - 3$$

Hence the answer is (D) $\lambda - 1$, where $\lambda = 3e^{2-t} - 3$

Exam Tip

If you have a basic knowledge of partial differential, you can differentiate the function with respect to u and get $\lambda - 1$ even if you do not have the concept of Optimal Control Theory. This leaves only choice (B) and (D) as an answer and increases your probability of selecting the correct choice. Therefore, do not hesitate to give some complex questions a try if you have extra time left after attempting the questions you can confidently answer.



Question 26

If the required reserve ratio (rr) is 10% and if the initial deposit is 100 Birr, what is the total amount of deposit created by the banking system?

- a. 1 Birr
- b. 10000 Birr
- c. 1000 Birr
- d. 10 Birr

Question Description	
Theme	Basic Macroeconomics
Course Name	Monetary Economics
Chapter	Chapter 4: The Money Supply Process

Solution

The multiple increase in deposits generated from an increase in the banking systems reserves is called the simple deposit multiplier. The formula for the multiple expansion of deposits can be written as follows:

$$\Delta D = \frac{1}{r} \times \Delta R$$

where

- ΔD is the change in total checkable deposits in the banking system.
- r is the required reserve ratio.
- ΔR is the change in reserves for the banking system.

$$\Delta D = \frac{100}{0.1} = 1000$$

Hence the answer is **(C)** 1,000 Birr.



Question 27

In economic analysis, the overvaluation of the exchange rate is corrected by employing _____

- a. Shadow exchange rate
- b. Social discount rate
- c. Shadow wage rate
- d. Marginal social cost

Question Description	
Theme	Basic Microeconomics
Course Name	Development Planning and Project Analysis II
Chapter	Chapter 3: Economic Analysis of Projects

Solution

Overvaluation of local currency is a common practice in developing countries. Currency overvaluation creates an apparent shortage of foreign exchange. This happens because at the overvalued exchange rate imports appear cheap relative to locally produced goods unless tariffs are imposed, demand for imports will rise. On the other hand, currency over valuation makes exporting as compared with supplying the local market, financially unattractive to producers. The price that exporters receive in local currency for a given dollar value of exports is reduced if the exchange rate is overvalued.

The official exchange rate is fixed at OER. This will result in excess demand for foreign exchange. In these circumstances the official exchange rate will understate the true value of foreign exchange to the country concerned. This is given by the shadow exchange rate, SER, the amount residents are willing to pay for the fixed quantity of foreign exchange available. Use of the OER in project appraisal will have the effect of undervaluing projects that produce exportable outputs and overvaluing those that use imported inputs. The overvaluation of the exchange rate must be corrected in an economic analysis. One method of doing this is to employ a shadow exchange rate to convert foreign prices into local currency.

Hence the answer is (A) shadow exchange rate.



Question 28

In the Tobin's q Theories of investment, Tobin's q is defined as:

- a. $q = \frac{\text{Market Value of Installed Capital}}{\text{Replacement Cost of Installed Capital}}$
- b. $q = \frac{\text{Market Value of Installed Capital}}{\text{Replacement Cost of Installed Capital}} \times \text{interest rate}$
- c. $q = \frac{\text{Replacement cost of Installed Capital}}{\text{Market value of Installed Capital}}$
- d. $q = \frac{\text{Replacement cost of Installed Capital}}{\text{Market value of Installed Capital}} \times \text{interest rate}$

Question Description	
Theme	Basic Macroeconomics
Course Name	Macroeconomics II
Chapter	Chapter 2: Theories of investment

Solution

Many economists see a link between fluctuations in investment and fluctuations in the stock market. stock prices reflect the incentives to invest. The Nobel Prize-winning economist James Tobin proposed that firms base their investment decisions on the following ratio, which is now called Tobin's q:

$$q = \frac{\text{Market value of Installed capital}}{\text{Replacement Cost of Installed Capital}}$$

The numerator of Tobin's q is the value of the economy's capital as determined by the stock market. The denominator is the price of that capital if it were purchased today.

Hence the answer is (A) $q = \frac{\text{Market value of Installed capital}}{\text{Replacement Cost of Installed Capital}}$

Additional Note

- If $q > 1$, the stock market values installed capital at more than its replacement cost. (Managers can raise the market value of their firm's stock by buying more capital)
- If $q < 1$, the stock market values capital at less than its replacement cost, (Firms do not replace capital as it wears out)



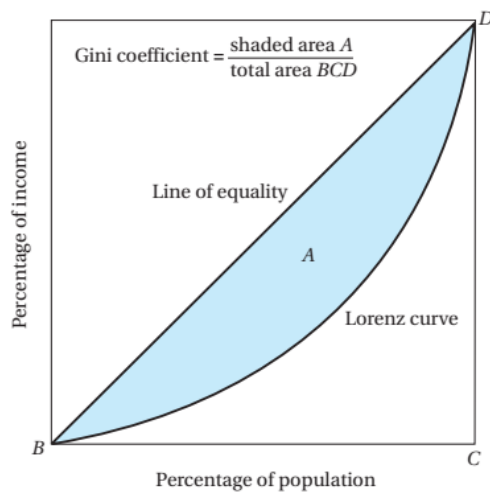
Question 29

In which one of the following cases of Lorenz curves is a comparison of the inequality levels of two countries valid?

- When the Lorenz curves of two countries are drawn in the same box but they do not cross one another
- The Lorenz curves lead to the Gini-coefficients is either near 1 or near zero
- When the Lorenz curves of two countries are drawn in the same box irrespective of whether they cross or not
- When the Gini-coefficient is computed without the Lorenz curves

Question Description	
Theme	Basic Macroeconomics
Course Name	Development economics I
Chapter	Chapter 5: Income Inequality, Poverty and Development

Solution



- The greater the curvature of the Lorenz line, the greater the relative degree of inequality.

- Gini coefficients are aggregate inequality measures and can vary anywhere from 0 (perfect equality) to 1 (perfect inequality).

As discussed in question 7, the Lorenz curve criterion states that we need more information to compare the level of inequality of countries if their Lorenz curves cross.

Hence the answer is **(A)** When the Lorenz curves of two countries are drawn in the same box but they do not cross one another.



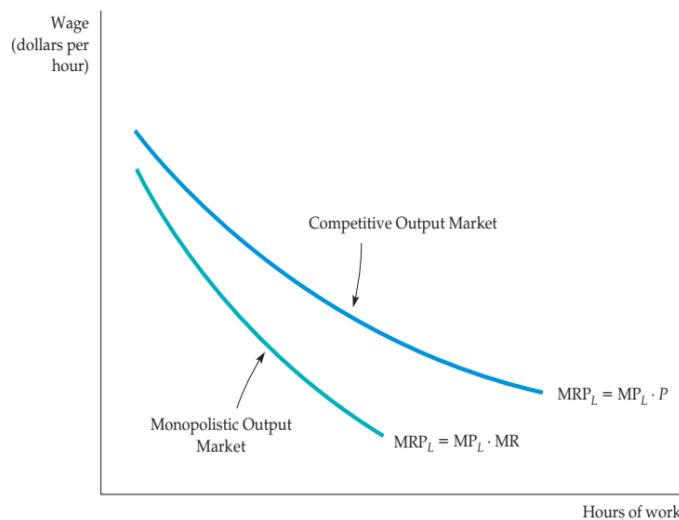
Question 30

Indicate the one that is responsible for the MRP curve to slope downward, if the firm is a perfect competitor in the product market

- a. Price continuously declines as the demand for resource increases
- b. Marginal physical product curve of the resource is downward sloping
- c. Both price and marginal product declines as the quantity of resource used rises
- d. Resource productivity rises as we use more of that resource

Question Description	
Theme	Basic Microeconomics
Course Name	Microeconomics II
Chapter	Chapter 4: Pricing of Factors of Production and Income distribution

Solution



Marginal revenue product is the additional revenue resulting from the sale of output created by the use of one additional unit of an input. In a competitive factor market in which the producer is a price taker, the buyer's demand for an input is given by the marginal revenue product curve. The MRP curve falls because the marginal product of labor falls as hours of work increase.

It is known that price is constant in a perfectly competitive market.

Hence the answer is **(B)** marginal physical product curve of the resource is downward sloping.



Question 31

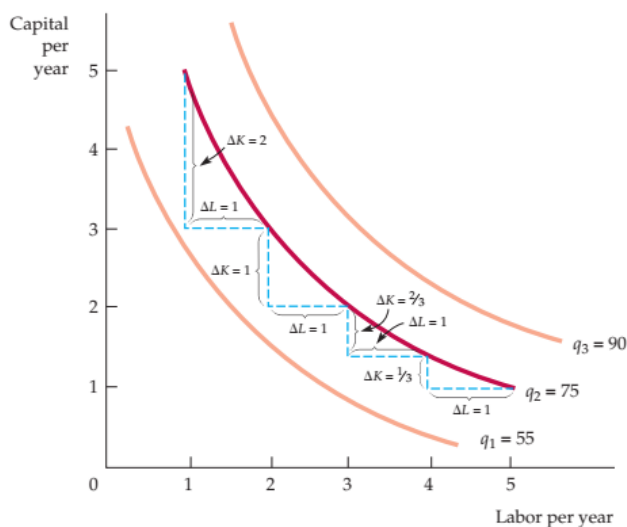
It expresses how much more output of x can be produced if the production of one unit of y is reduced by a unit

- a. Marginal rate of product transformation
- b. Competitive exchange
- c. Production Possibility Frontier
- d. The first theorem of welfare economics

Question Description	
Theme	Basic Microeconomics
Course Name	Microeconomics I
Chapter	Chapter 3: Theory of Production

Solution

Isoquant Curve showing all possible combinations of inputs that yield the same output. The slope of each isoquant indicates how the quantity of one input can be traded off against the quantity of the other, while output is held constant. When the negative sign is removed, we call the slope the marginal rate of technical substitution (MRTS). The marginal rate of technical substitution of labor for capital is the amount by which the input of capital can be reduced when one extra unit of labor is used, so that output remains constant.



Like indifference curves, isoquants are downward sloping and convex. The slope of the isoquant at any point measures the marginal rate of technical substitution—the ability of the firm to replace capital with labor while maintaining the same level of output. On isoquant q_2 , the MRTS falls from 2 to 1 to $2/3$ to $1/3$.

Hence the answer is (A) Marginal rate of product transformation.



Additional Note

The production possibilities frontier shows the various combinations of food and clothing that can be produced with fixed inputs of labor and capital, holding technology constant. Marginal rate of transformation (MRT) is the amount of one good that must be given up to produce one additional unit of a second good.

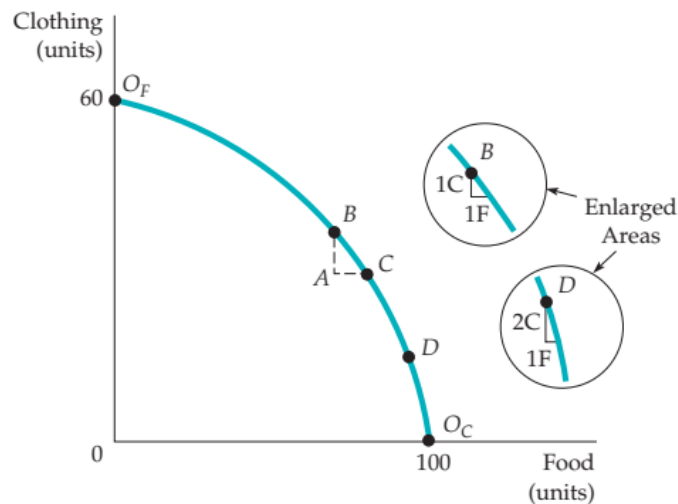


Figure: Production Possibilities Frontier

The MRT measures how much clothing must be given up to produce one additional unit of food. For example, the enlarged areas of the figure to the left, show that at B on the frontier, the MRT is 1 because 1 unit of clothing must be given up to obtain 1 additional unit of food. At D, however, the MRT is 2 because 2 units of clothing must be given up to obtain 1 more unit of food.



Question 32

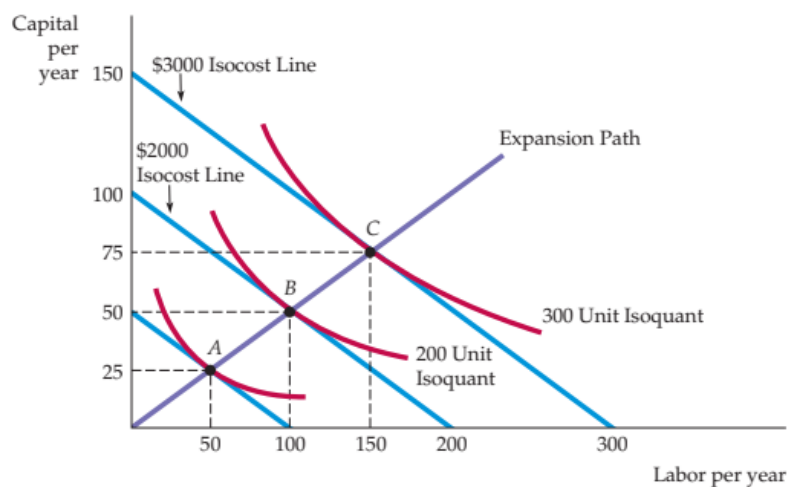
It is the locus of points of producer equilibrium resulting from changes in total monetary outlay, while keeping factor prices constant.

- a. Expansion path
- b. Production function
- c. Isocost
- d. Isoquant

Question Description	
Theme	Basic Microeconomics
Course Name	Microeconomics I
Chapter	Chapter 3: Theory of Production

Solution

The curve passing through the points of tangency between the firm's isocost lines and its isoquants is its expansion path. The expansion path describes the combinations of labor and capital that the firm will choose to minimize costs at each output level. As long as the use of both labor and capital increases with output, the curve will be upward sloping.



The expansion path (from the origin through points A, B, and C) shown in the figure to the left illustrates the lowest-cost combinations of labor and capital that can be used to produce each level of output in the long run.

Hence the answer is **(A)** Expansion Path.



Question 33

Non-discrimination as one of the key principles of the GATT, has two applications which are ____

- Un-favoured nation (non-discriminatory treatment of similar goods coming from two different countries) and international Treatment (non-discriminatory treatment of similar domestic and foreign goods)
- Most favoured nation (discriminatory treatment of similar goods coming from two different countries) and National Treatment (discriminatory treatment of similar domestic and foreign goods)
- Most favoured nation (non-discriminatory treatment of similar goods coming from two different countries) and National Treatment (non-discriminatory treatment of similar domestic and foreign goods)
- Most favoured nation (non-discriminatory treatment of similar domestic and foreign goods) and National Treatment (non-discriminatory treatment of similar goods coming from two different countries)

Question Description	
Theme	Basic Macroeconomics
Course Name	International Economics I
Chapter	Chapter 5: International Trade Institutions and Regional Economic Arrangements

Solution GATT rested on three basic principles:

1. Nondiscrimination.

The principle of National Treatment: requires that goods, once lawfully imported, will be treated in the same manner as domestic goods.

Most favored nation principle: the state parties must apply the same duties and charges on the import of goods between GAAT parties, regardless of origin.

2. Elimination of nontariff trade barriers (such as quotas), except for agricultural products and for nations in balance-of-payments difficulties.

3. Consultation among nations in solving trade disputes within the GATT framework

Hence the answer is (C) Most favored nation (non-discriminatory treatment of similar goods coming from two different countries) and National Treatment (non-discriminatory treatment of similar goods coming from two different countries)



Question 34

One of the following best describes the current state of international trade

- a. the flow of trade in goods and services is next to that of capital flows
- b. Capital flows more freely across national boundaries than people.
- c. free international flow of people exceeds the free trade in goods and services,
- d. International flow of people is freer than international flow capital

Question Description	
Theme	Basic Macroeconomics
Course Name	International Economics I
Chapter	Chapter 1: Introduction

Solution

Interdependence in the world economy is reflected in the flow of goods, services, labor, and capital across national boundaries.

Most nations impose restrictions on immigration to reduce the inflow of low-skilled people. Migration is generally more restricted and regulated than the international flow of goods, services, and capital.

In general, capital flows more freely across national boundaries than people. Financial or portfolio capital (bank loans and bonds) generally move to nations and markets where interest rates are higher, and foreign direct investments in plants and firms flows to nations where expected profits are higher. This leads to the more efficient use of capital and generally benefits both lenders and borrowers.

Hence the answer is **(B)** capital flows more freely across national boundaries than people.



Question 35

One of the following is an expression that correctly distinguishes Harod-Domar Model of growth from Solow model of growth.

- For Solow growth model labour force growth is not important in as is assumed to be abundant in a developing economy while labour is important for Harod-Domar growth model.
- In Harod –Domar model, net additions to the capital stock will bring about corresponding increases in the flow of national output, while in Solow Model, net additions to capital have diminishing returns
- In Solow model, net additions to the capital stock will bring about corresponding increases in the flow of national output, while net additions to capital have diminishing returns in Harod-Domar Model
- Technological progress can be expressed in the Solow growth model context as a decrease in the required capital-output ratio while technology is exogenously given in Harod-Domar model

Question Description	
Theme	Basic Macroeconomics
Course Name	Development economics I
Chapter	Chapter 3: Growth Models and Theories of development

Solution

The key modification from the Harrod-Domar (or AK) growth model is that the Solow model allows for substitution between capital and labor. In the process, it assumes that there are diminishing returns to the use of these inputs.

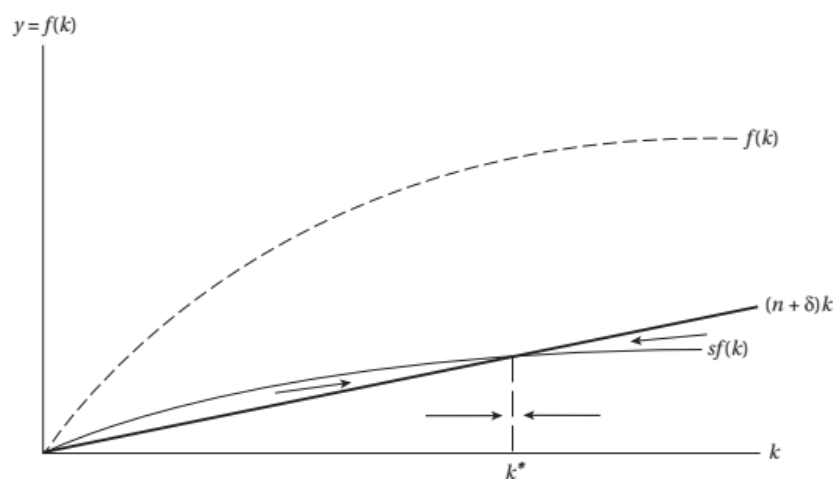


Figure: Equilibrium in the Solow growth model



In the Solow growth model graphed above, The concave shape of $f(k)$ —that is, increasing at a decreasing rate—reflects diminishing returns to capital per worker. In the Harrod-Domar model, this would instead be a straight, upward-sloping line.

In the context of the Harrod-Domar model, labor force growth is not described explicitly. This is because labor is assumed to be abundant in a developing-country context and can be hired as needed in a given proportion to capital investments (this assumption is not always valid). In a general way, technological progress can be expressed in the Harrod-Domar context as a decrease in the required capital-output ratio, giving more growth for a given level of investment.

Hence the answer is (B) in Harrod-Domar model, net additions to the capital stock will bring about corresponding increases in the flow of national output, while in Solow Model, net additions to capital have diminishing returns.



Question 36

One of the following is the most important “power” of the WTO?

- a. The ability to adjudicate disputes between member countries regarding compliance with the Agreements
- b. using coercion rather resorting to soliciting compliance or appeal to suspension of concessions
- c. To sustain the complete set of Uruguay Round agreements by applying trade sanctions
- d. The ability to enforce free trade among member countries.

Question Description	
Theme	Basic Macroeconomics
Course Name	International Economics I
Chapter	Chapter 5: International Trade Institutions and Regional Economic Arrangements

Solution

- The WTO’s main purpose is to monitor the trade liberalization agreements reached by GATT member countries in the Uruguay Round.
- The most important “power” of the WTO is its ability to adjudicate disputes between member countries regarding compliance with the Agreements.
- Dispute resolution is conducted by the Dispute Settlement Body (DSB), which includes one representative from each WTO government.
- The four main steps to a WTO dispute case are
 1. consultations,
 2. panel formation,
 3. appeals, and
 4. resolution.

Hence the answer is **(A)** The ability to adjudicate disputes between member countries regarding compliance with the Agreements.



Question 37

Place the following in the correct order:

- I. Identifying the Key Factors
- II. Collecting Data and Information
- III. Determining the Flow of Information
- IV. Developing a Measuring System
- V. Preparing Reports Based on Data

- a. II, I, III, IV, V
- b. I, IV, II, V, III
- c. IV, I, II, III, V
- d. I, IV, V, III, II

Question Description	
Theme	Uncategorized
Course Name	Uncategorized
Chapter	Uncategorized



Question 38

Pursuing expansionary monetary policy in conjunction with large budgetary deficits is expected to result in _____.

- a. Interest rates that promote the expansion of output while increasing demand
- b. Promote industrial output growth but reduces demand
- c. Positive real interest rates where inflation rates fall below nominal interest levels.
- d. Negative real interest rates where inflation rates exceed nominal interest levels.

Question Description	
Theme	Basic Macroeconomics
Course Name	Development economics II
Chapter	Chapter 6: Foreign aid, debt, financial reform and development

Solution

Monetary policy: Activities of a central bank designed to influence financial variables such as the money supply and interest rates.

In developing nations, investment decisions are often not very sensitive to interest-rate movements. Moreover, a number of larger countries in Latin America (e.g., Brazil and Argentina) have in the past followed a policy of inflation-financed industrial growth, in which expansionary monetary policy in conjunction with large budgetary deficits resulted in negative real interest rates (inflation rates exceeding nominal interest levels). The basic idea was that artificially low rates would encourage investment, finance the fiscal deficit, and Currency substitution The use of foreign currency (e.g., U.S. dollars) as a medium of exchange in place of or along with the local currency (e.g., Mexican pesos).

Hence the answer is **(D)** Negative real interest rates where inflation rates fall below nominal interest rates.



Question 39

Regulation of financial institutions is conducted by

- a. Insurance companies
- b. Central banks
- c. Commercial banks
- d. Credit associations

Question Description	
Theme	Basic Macroeconomics
Course Name	Monetary Economics
Chapter	Chapter 5: Central Banking and Monetary Policy

Solution

Functions of a central bank might include:

- Monopoly on the issue of banknotes
- The government's banker
- The bankers' bank ("lender of last resort")
- Manages the country's foreign exchange and gold reserves and the government's stock register;
- Regulation and supervision of the banking industry:
- Setting the official interest rate - used to manage both inflation and the country's exchange rate - and ensuring that this rate takes effect via a variety of policy mechanisms

Hence the answer is **(B)** Central Banks.



Question 40

Suppose a given economy is represented by the following equations:

$Y = C + I + G + X - M$, $C = 400 + 0.8Y_d$, 10 million, lump sum tax (T) = 15 million, export (X) = 120 million, import (M) = $600 + 0.2Y$ and investment (I) = 26 million. What are the equilibrium national income (in million) and the import multiplier, respectively?

- a. 100 and -2.5
- b. 150 and -0.25
- c. 200 and -0.75
- d. 120 and -0.5

Question Description	
Theme	Quantitative Methods in Economics with application
Course Name	Mathematical Economics
Chapter	Chapter 3: Comparative Statistic Analysis

Solution

There seems to be a missing information from this question. Let us assume the government expenditure to be 106 million and see how we can solve the question.

$$Y = C + I + G + X - M$$

$$Y = 400 + 0.8Y_d + 26 + 106 + 120 - 600 - 0.2Y$$

But disposable income is income after taxation $Y_d = Y - T$

$$Y = 400 + 0.8(Y - 15) + 26 + 106 + 120 - 600 - 0.2Y$$

$$Y = 400 + 0.8Y - 12 + 26 + 106 + 120 - 600 - 0.2Y$$

$$0.4Y = 40$$

$$Y = 100$$

The import multiplier is given by

$$\text{Import multiplier} = -\frac{1}{1 - \text{MPC} + \text{MPI}} = -\frac{1}{1 - 0.8 + 0.2} = -\frac{1}{0.4} = -2.5$$

Hence the answer would have been (A) 100 and -2.5 had $G = 106$ million.



Question 41

Suppose the Ethiopian population is estimated to be 110 million in the year 2012 E.C. the number of children below the age of 14 is 35 million while the number of the elder ones above the age of 65 is 15 million. If the total number of economically inactive population is 10 million whereas those people who are employed are 30 million. Then what are the total number of unemployed people and the unemployment rate, respectively?

- a. 40 million and 20%
- b. 20 million and 20%
- c. 40 million and 40%
- d. 20 million and 40%

Question Description	
Theme	Basic Macroeconomics
Course Name	Macroeconomics II
Chapter	Chapter 4: The Labor Market

Solution

A person to be categorized as unemployed, two conditions must be fulfilled

- The person is without a job and able to work.
- The person wants to have a job and is willing to work at the current market wage rate.

$$\text{Labor force} = 110 - 35 - 15 - 10 = 50 \text{ million}$$

$$\text{Total number of unemployed} = \text{LF} - \text{Total number of employed}$$

$$\text{Total number of unemployed} = 50 - 30 = 20 \text{ million}$$

$$\text{Unemployment rate} = \frac{\text{Total number of unemployed}}{\text{Labor force}} \times 100\%$$

$$\text{Unemployment rate} = \frac{20}{50} \times 100\% = 40\%$$

Hence the answer is **(D)** 20 million and 40%.



Question 42

Suppose the Marshall – Learner condition does not hold. A real depreciation will tend to cause

- a. a reduction in net export (NX) only
- b. An increase in income and export
- c. a reduction in income (Y) only
- d. A decrease in both income and export

Question Description	
Theme	Basic Macroeconomics
Course Name	International Economics II
Chapter	Chapter 3: Balance of Payments

Solution

Marshall–Lerner condition is the condition under which a real depreciation leads to an increase in net exports. If the Marshall-Lerner condition is not satisfied, then a depreciation (an appreciation) of a country's currency will lead to a more negative (more positive) trade balance for the country.

Hence the answer is **(D)** a reduction in income and export.



Question 43

Suppose there are two strategies open to each competitor-low price & high price strategies. The payoff table of the two firms is shown as follows. What is the logical outcome the game for the two players in the absence of collusion?

		Firm B's Strategies	
		Low Price	High Price
Firm A's Strategies	Low Price	(1000,1000)	(3000,-1000)
	High Price	(-1000,3000)	(2000,2000)

- Firm A should go for strategy of low price and that of B for high price
- Both firm A and B should go for strategy of low price
- Both firm and B should go for strategy of high price
- Firm B should go for strategy of low price and that of A for high price

Question Description	
Theme	Basic Microeconomics
Course Name	Microeconomics II
Chapter	Chapter 3: Game Theory

Solution

In the absence of collision,

- Player A plays low price if player B plays low price. Player A also plays low price if Player B plays high price.
- Player B plays low price if player A plays low price. Player B also plays low price if Player A plays high price.

Both players play their dominant strategies and the outcome of the game will be (Low price, Low price)

Hence the answer is **(B)** Both firm A and B should go for strategy low price.

Additional Note

Note that the Nash Equilibrium is not the socially optimal strategy. If the firms colluded, they could have both earned better payoffs i.e. (2000,2000)



Question 44

The Centralized cartel _____

- a. Main aim is to maximize each firm's independent profit by equating $MR = P$
- b. decide only total quantity and the optimal price and the allocation of production and distribution is to be determined by each firm
- c. Allocates total production among firms using the following rule: $MC = MR_1 = MR_2$
- d. Is formed by firms with the aim of reducing uncertainty arising from their mutual interdependence

Question Description	
Theme	Basic Microeconomics
Course Name	Microeconomics II
Chapter	Chapter 2: Oligopoly

Solution

Centralized Cartel (Cartel aiming at joint profit maximization) is formed as its name indicates with the objective of maximizing the total industry's profit. In order to achieve this objective, member firms appoint a central agency to which they delegate the authority to decide not only the total quantity and price at which the industry's profit maximized but also allocation of output and profit among the member of cartel.

Cartels imply direct (although secret) agreements among the competing oligopolist with the aim of reducing the uncertainty arising from their mutual interdependence.

The mathematical presentation of the cartel model which aims at joint profit maximization is identical to that of the multiplant monopolist i.e.

$$MR = MC_1 = MC_2$$

Hence the answer is **(D)** is formed by firms with the aim of reducing uncertainty arising from their mutual interdependence.



Question 45

The collapse of the Bretton Woods system was partly due to

- a. the unbalanced macroeconomic power of the United States, which allowed it to generate global inflation
- b. the skewed macroeconomic power of the United States, which allowed it to generate global deflation
- c. the fact that all countries have achieved internal and external balances.
- d. the balanced macroeconomic power of the United States, which allowed it to generate global inflation

Question Description	
Theme	Basic Macroeconomics
Course Name	International Economics II
Chapter	Chapter 5: International Monetary System and Key International Financial Institutions

Solution

While the immediate cause of the collapse of the Bretton Woods system was the huge balance-of-payments deficits of the United States in 1970 and 1971, the fundamental cause is to be found in the interrelated problems of liquidity, adjustment, and confidence.

Hence the answer is **(A)** the unbalanced macroeconomic power of the United States, which allowed it to generate global inflation.



Question 46

The common features of developing countries in the classification of the United Nations Development Programme (UNDP) are -----

- a. Adverse geography and Underdeveloped financial and other markets
- b. Lingering colonial impacts such as poor institutions and often external dependence.
- c. Low standard of living and low health and education attainments
- d. Slow industrialization and urbanization

Question Description	
Theme	Basic Macroeconomics
Course Name	Development economics I
Chapter	Chapter 2: Structural Features and common characteristics of developing countries.

Solution

According to Todaro, there are ten important features that developing countries tend to have in common, on average, in comparison with the developed world. These areas are the following:

1. Lower levels of living and productivity
2. Lower levels of human capital
3. Higher levels of inequality and absolute poverty
4. Higher population growth rates
5. Greater social fractionalization
6. Larger rural populations but rapid rural-to-urban migration
7. Lower levels of industrialization
8. Adverse geography
9. Underdeveloped financial and other markets
10. Lingering colonial impacts such as poor institutions and often external dependence.

The UNDP classifies countries according to their level of human development, including health and education attainment as Low, Medium, High and Very High.

Hence the answer is (C) low standard of living, low health and educational attainments.



Question 47

The cost function of a particular firm is given as, $C = 10q - 2q^2 + q^3$. Find the output level at which the average cost is minimized and the minimum level of output, respectively.

- a. 1 and 6
- b. 1 and 9
- c. 2 and 9
- d. 2 and 6

Question Description	
Theme	Quantitative Methods in Economics with application
Course Name	Mathematical Economics
Chapter	Chapter 1: Derivatives in Use

Solution

$$AC = \frac{TC}{q} = \frac{10q - 2q^2 + q^3}{q} = 10 - 2q + q^2$$

We need to find the derivative of the function to determine the level of output where average cost is minimized.

$$\frac{dAC}{dx} = 2q - 2 = 0$$

$$q = 1$$

Average cost at the minimum level of output = $10 - 2(1) + (1)^2 = 9$

Total cost at the minimum level of output = $10(1) - 2(1)^2 + (1)^3 = 9$

Hence the answer would have been **(B)** 1 and 9 had the question asked the level of output where average cost is minimized and the average cost at the minimum level of output.



Question 48

The ex-post analysis/evaluation is inadequate for project control because

- a. it doesn't enable the firm in identifying individual with superior abilities in planning and forecasting
- b. it is a forward looking rather than backward looking
- c. it doesn't provide a documented log of experience that may be valuable in improving future decision making
- d. it does not use the data effectively to provide integrated control

Question Description	
Theme	Basic Microeconomics
Course Name	Development Planning and Project Analysis II
Chapter	Chapter 2: Financial Analysis and Appraisal of Projects

Solution

The ex-post analysis/evaluation is inadequate for project control because it is a backward looking rather than forward looking and it does not use the data effectively to provide integrated control. Ex-post/post audit action of a project control:

- provides a documented log of experience that may be valuable in improving future decision making
- enables the firm in identifying individual with superior abilities in planning and forecasting
- helps in discovering systematic biases in judgment induce healthy caution among project sponsors, and
- serves as a useful training ground for promising executives who need broader business experience and exposure.

Hence the answer is **(D)** it does not use the data effectively to provide integrated control.



Question 49

The interest parity condition written above now tells us that

- If people expect relative PPP to hold, the difference between the interest rates offered by US dollar and Ethiopian Birr deposits will equal the difference between the inflation rates expected, over the relevant horizon, in the United States and in Ethiopia
- If people expect relative PPP to hold, the difference between the interest rates offered by US dollar and Ethiopian Birr deposits will equal the difference between the inflation rates expected, over the relevant horizon, in the United States but not in Ethiopia in Ethiopia
- If people expect relative PPP to hold, the difference between the interest rates offered by US dollar and Ethiopian Birr deposits will equal the difference between the inflation rates expected, over the relevant horizon, and in Ethiopia but not the in the United States
- If people do not expect relative PPP to hold, the difference between the interest rates offered by US dollar and Ethiopian Birr deposits will equal the difference between the inflation rates expected, over the relevant horizon, in the United States and in Ethiopia

Question Description	
Theme	Basic Macroeconomics
Course Name	International Economics II
Chapter	Chapter 2: Theories of Exchange rate determination

Solution

The interest rate parity condition is missing in the exam question. I believe this is the condition the exam makers intended to provide.

$$R_{\$} = R_{ETB} + (E_{\$/ETB}^e - E_{\$/ETB})/E_{\$/ETB}$$

According to relative PPP, the percentage change in the dollar/Ethiopian Birr rate over the next year, say, will equal the difference between the inflation rates of the United States and Ethiopian Birr over that year. Since people understand this relationship, however, it must also be true that they expect the percentage exchange rate change to equal the U.S.– Ethiopian Birr inflation difference. The interest parity condition written above now tells us the following: If people expect relative PPP to hold, the difference between the interest rates offered by dollar and Ethiopian Birr deposits will equal the



difference between the inflation rates expected, over the relevant horizon, in the United States and in Ethiopia.

Hence the answer is (A) If people expect relative PPP to hold, the difference between the interest rates offered by dollar and Ethiopian Birr deposits will equal the difference between the inflation rates expected, over the relevant horizon, in the United States and in Ethiopia.



Question 50

The most important function of central bank is

- a. controlling Money supply
- b. job creation
- c. promote export
- d. promote import

Question Description	
Theme	Basic Macroeconomics
Course Name	Monetary Economics
Chapter	Chapter 5: Central Banking and Monetary Policy

Solution

Functions of a central bank might include:

- Monopoly on the issue of banknotes
- The government's banker
- The bankers' bank ("lender of last resort")
- Manages the country's foreign exchange and gold reserves and the government's stock register;
- Regulation and supervision of the banking industry:
- Setting the official interest rate - used to manage both inflation and the country's exchange rate - and ensuring that this rate takes effect via a variety of policy mechanisms.

Hence the answer is (A) controlling money supply.



Question 51

The Mundell–Fleming model shows that

- a. Fiscal policy does not influence aggregate income under fixed exchange rates.
- b. Monetary policy does influence aggregate income under fixed exchange rates
- c. Fiscal policy does not influence aggregate income under floating exchange rates
- d. Monetary policy does not influence aggregate income under floating exchange rates.

Question Description	
Theme	Basic Macroeconomics
Course Name	Macroeconomics I
Chapter	Chapter 4: The Mundell-Fleming Model

Solution

The Mundell–Fleming Model: Summary of Policy Effect

Policy	EXCHANGE-RATE REGIME					
	FLOATING			FIXED		
	IMPACT ON:					
	Y	e	NX	Y	e	NX
Fiscal expansion	0	↑	↓	↑	0	0
Monetary expansion	↑	↓	↑	0	0	0
Import restriction	0	↑	0	↑	0	↑

Note: This table shows the direction of impact of various economic policies on income Y , the exchange rate e , and the trade balance NX . A “↑” indicates that the variable increases; a “↓” indicates that it decreases; a “0” indicates no effect. Remember that the exchange rate is defined as the amount of foreign currency per unit of domestic currency (for example, 100 yen per dollar).

Hence the answer is **(C)** Fiscal policy does not influence aggregate income under floating exchange rates.



Question 52

The relationship between, labor force participation rate(LFPR in percent), labor force(LF) and adult population (AP) can be formulated as;

- a. $LFPR = \frac{AP}{LF} \times 100$
- b. $LFPR = \frac{AP}{AP+LF} \times 100$
- c. $LFPR = \frac{LF}{AP} \times 100$
- d. $LFPR = \frac{LF}{AP+LF} \times 100$

Question Description	
Theme	Basic Macroeconomics
Course Name	Macroeconomics II
Chapter	Chapter 4: The Labor Market

Solution

$$\text{Labor force participation rate} = \frac{\text{Labor Force}}{\text{Adult Population}} \times 100\%$$

Hence the answer (C) is $LFPR = \frac{LF}{AP} \times 100$



Question 53

The procedure of estimating structural parameters from the coefficients of the reduced-form equations are called

- a. indirect least squares (ILS)
- b. direct least squares (DLS)
- c. Error correction
- d. Vector auto regression

Question Description	
Theme	Quantitative Methods in Economics with application
Course Name	Econometrics II
Chapter	Chapter 2: Introduction to Simultaneous Equation Models

Solution

- For a just or exactly identified structural equation, the method of obtaining the estimates of the structural coefficients from the OLS estimates of the reduced-form coefficients is known as the method of indirect least squares (ILS), and the estimates thus obtained are known as the indirect least-squares estimates.
- The error correction mechanism (ECM) developed by Engle and Granger is a means of reconciling the short-run behavior of an economic variable with its long-run behavior.
- Broadly speaking, there are five approaches to economic forecasting based on time series data:
 1. exponential smoothing methods,
 2. single-equation regression models,
 3. simultaneous-equation regression models,
 4. autoregressive integrated moving average (ARIMA) models, and
 5. vector autoregression (VAR) models.

Hence the answer is (A) is indirect least squares



Additional Note

The method of 2SLS (Two-Stage Least Squares) is especially designed for overidentified equations, although it can also be applied to exactly identified equations. But then the results of 2SLS and ILS are identical. The basic idea behind 2SLS is to replace the (stochastic) endogenous explanatory variable by a linear combination of the predetermined variables in the model and use this combination as the explanatory variable in lieu of the original endogenous variable.



Question 54

The relationship between Monetary base or the high-powered money (H) is given in terms of currency in the hands of the public (C) and Bank Reserves (R) as

- a. $H = C + R$
- b. $H = C - R$
- c. $H = R - C$
- d. $H = C \times R$

Question Description	
Theme	Basic Macroeconomics
Course Name	Monetary Economics
Chapter	Chapter 4: The money supply process

Solution

The monetary base (also called high-powered money) equals currency in circulation C plus the total reserves in banking system. The monetary base MB can be expressed as

$$MB = C + R$$

The Federal Reserve exercises control over the monetary base through its purchases or sale of government securities in the open market, called open market operations, and through the its extension of discount loans to banks.

Hence the answer is **(A)** $H = C + R$



Question 55

The relationship between real exchange rate, nominal exchange rate domestic price and foreign price can be summarized as

- $RealExchangeRate = \frac{Nominal\ Exchange\ Rate \times Domestic\ Price}{Foreign\ price}$
- $RealExchangeRate = \frac{Nominal\ Exchange\ Rate \times Domestic\ Price}{Foreign\ interest\ rate}$
- $Real\ Exchange\ rate = \frac{Nominal\ Exchange\ Rate \times Foreign\ Price}{Domestic\ Price}$
- $RealExchangeRate = \frac{Nominal\ Exchange\ Rate \times Domestic\ Price}{Domestic\ interest\ rate}$

Question Description	
Theme	Basic Macroeconomics
Course Name	International Economics II
Chapter	Chapter 1: The Foreign Exchange Market

Solution

The real exchange rate (RER) between two currencies is the product of the nominal exchange rate (the dollar cost of a euro, for example) and the ratio of prices between the two countries.

The core equation is $RER = e \times \frac{P^*}{P}$

where

- e is the nominal dollar/euro exchange rate,
- P^* is the average price of a good in the euro area, and
- P is the average price of the good in the United States.

For example, if $e = 1.36$. If the German price of a certain commodity is 2.5 euros and the U.S. price is \$3.40, then $(1.36) \times (2.5) \div 3.40$ yields an RER of 1. But if the German price were 3 euros and the U.S. price \$3.40, the RER would be $1.36 \times 3 \div 3.40$, for an RER of 1.2.

Hence the answer is (A)

$$Real\ Exchange\ Rate = \frac{Nominal\ Exchange\ Rate \times Domestic\ Price}{Foreign\ Price}$$



Question 56

The stream of financial costs and benefits generated by the project over its economic life using shadow prices to meet equity objective is called _____?

- a. Financial cash flow
- b. Economic net benefit stream
- c. Social net benefit stream
- d. Social impact analysis

Question Description	
Theme	Basic Microeconomics
Course Name	Development Planning and Project Analysis II
Chapter	Chapter 3: Economic Analysis of Projects

Solution

Whether one is calculating financial or economic rate of return, a key part of the analysis will be to work out the actual flows of income and expenditure. The financial cash flow of a project is the stream of financial costs and benefits, or expenditures and receipts that will be generated by the project over its economic life, and will not be produced in its absence. Note that when one is working with market prices, the cash flow stream is referred to as financial Cash Flow. When these prices are adjusted to reflect national efficiency and equity objective (i.e. when shadow prices are used) it is referred to as Economic Net Benefit Stream and Social Net Benefit Stream, respectively.

Hence the answer is (C) Social Net Benefit Stream.



Question 57

The use of non-stationary variables in OLS can lead to

- a. error correction model
- b. regression with exact identification
- c. cross sectional regression
- d. spurious regressions

Question Description	
Theme	Quantitative Methods in Economics with application
Course Name	Econometrics II
Chapter	Chapter 3: Introduction to Basic Regression Analysis with Time Series Data

Solution

Nonstationarity can lead to spurious regression, an apparent relationship between variables that are, in reality not related.

In regressing a time series variable on another time series variable(s), one often obtains a very high R^2 (in excess of 0.9) even though there is no meaningful relationship between the two variables. Sometimes we expect no relationship between two variables, yet a regression of one on the other variable often shows a significant relationship. This situation exemplifies the problem of spurious, or nonsense, regression. One way to guard against it is to find out if the time series are cointegrated.

Hence the answer is **(D)** spurious regression.

Additional Note

Cointegration means that despite being individually nonstationary, a linear combination of two or more time series can be stationary. The Engle–Granger (EG) and the augmented Engle–Granger (AEG) tests can be used to find out if two or more time series are cointegrated.

Cointegration of two (or more) time series suggests that there is a long-run, or equilibrium, relationship between them.



Question 58

The value of each possible outcome times the probability of that outcome is ____

- a. Variance
- b. Expected value
- c. Expected utility
- d. Risk premium

Question Description	
Theme	Basic Microeconomics
Course Name	Microeconomics I
Chapter	Chapter 2: Choice involving risk and uncertainty

Solution

Here are the definition of the terms in the choices

- Expected value: probability weighted average of the payoffs associated with all possible outcomes.
- Variability: Extent to which possible outcomes of an uncertain event differ
- Expected utility: Sum of the utilities associated with all possible outcomes, weighted by the probability that each outcome will occur.
- Risk premium: Maximum amount of money that a risk averse person will pay to avoid taking a risk.

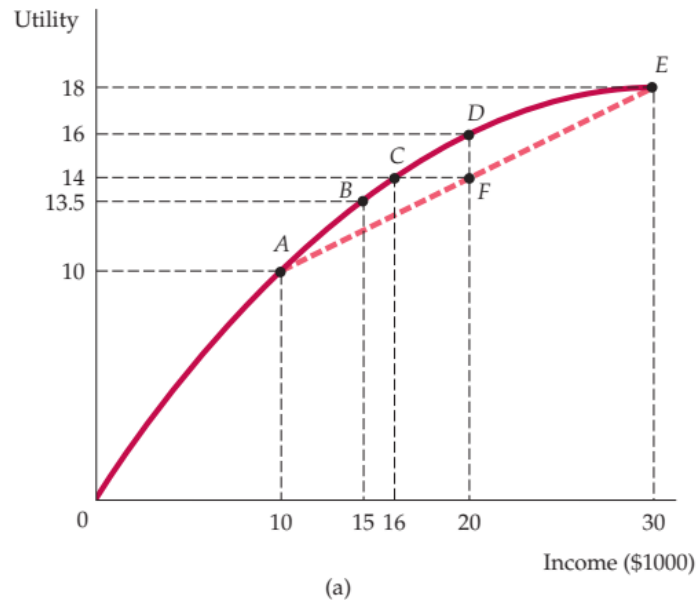
Hence the answer is **(B)** Expected value.

Additional Information

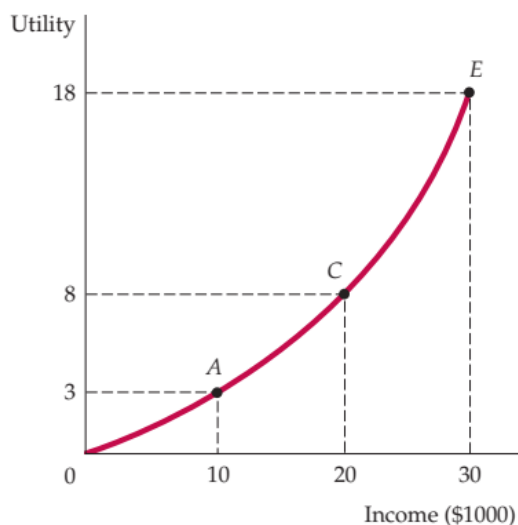
People differ in their willingness to bear risk. Some are risk averse, some risk loving, and some risk neutral.

- risk averse: Condition of preferring a certain income to a risky income with the same expected value.
- risk neutral: Condition of being indifferent between a certain income and an uncertain income with the same expected value.
- risk loving: Condition of preferring a risky income to a certain income with the same expected value.

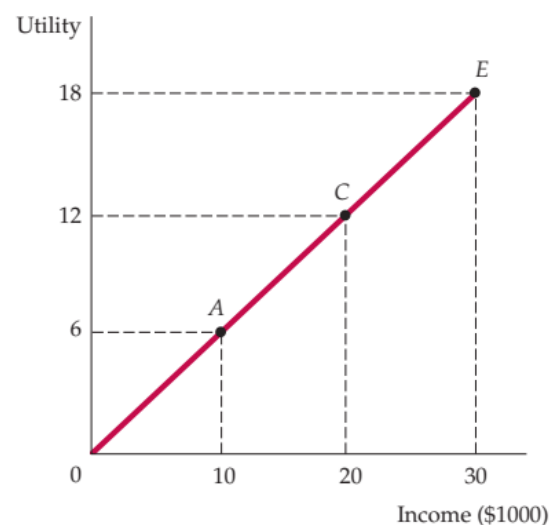




(a)



(b)



(c)

In (a), a consumer's marginal utility diminishes as income increases. The consumer is risk averse because she would prefer a certain income of \$20,000 (with a utility of 16) to a gamble with a .5 probability of \$10,000 and a .5 probability of \$30,000 (and expected utility of 14). In (b), the consumer is risk loving: She would prefer the same gamble (with expected utility of 10.5) to the certain income (with a utility of 8). Finally, the consumer in (c) is risk neutral and indifferent between certain and uncertain events with the same expected income.



Question 59

What are the components of official development assistance (ODA)?

- a. Commercial loans, concessional loans, purchase of bonds and multilateral flows
- b. Special drawing rights, commercial loans and multilateral grants
- c. Technical assistance, multilateral flows, preferential trade and long term loans
- d. Bilateral grants, concessional loans, technical assistance, and multilateral flows

Question Description	
Theme	Basic Macroeconomics
Course Name	Development economics II
Chapter	Chapter 6: Foreign aid, debt, financial reform and development

Solution

Official development assistance (ODA): Net disbursements of loans or grants made on concessional terms by official agencies, historically by high-income member countries of the Organization for Economic Cooperation and Development (OECD).

The money volume of official development assistance (ODA) includes bilateral grants, concessional loans, and technical assistance, as well as multilateral flows.

Hence the answer is **(D)** Bilateral grants, concessional loans, and technical assistance, as well as multilateral flows.



Question 60

What is meant by $u_i = N(0, \delta^2)$?

- a. The error term is normally distributed
- b. Variance of the error term is constant
- c. There is no autocorrelation
- d. The independent variables are not related

Question Description	
Theme	Quantitative Methods in Economics with application
Course Name	Econometrics I
Chapter	Chapter 2: Simple Linear Regression

Solution

The classical normal linear regression model assumes that each u_i is distributed normally with

$$\text{Mean: } E(u_i) = 0$$

$$\text{Variance: } E[u_i - E(u_i)]^2 = E(u_i^2) = \sigma^2$$

$$\text{cov}(u_i, u_j): E\{[(u_i - E(u_i))][u_j - E(u_j)]\} = E(u_i u_j) = 0 \quad i \neq j$$

The assumptions given above can be more compactly stated as

$$u_i \sim N(0, \delta^2)$$

where the symbol $\sim$ means distributed as and N stands for the normal distribution, the terms in the parentheses representing the two parameters of the normal distribution, namely, the mean and the variance.

Hence the correct answer is (A) the error term is normally distributed.

Note

I am assuming that the exam makers intended to ask about the meaning of the whole expression as opposed to the parts. Please note that the statement in choice (B) can also be inferred from the expression.



Question 61

What is the Hausmann test in in panel data econometrics

- a. Identification test
- b. Test of cointegration
- c. Test of structural break
- d. Specification test

Question Description	
Theme	Quantitative Methods in Economics with application
Course Name	Econometrics II
Chapter	Chapter 4: Introduction to Panel Data Regression Models

Solution

Panel regression models are based on panel data. Panel data consist of observations on the same cross-sectional, or individual, units over several time periods.

Despite their substantial advantages, panel data pose several estimation and inference problems. There are several estimation techniques to address one or more of these problems. The two most prominent are

1. the fixed effects model (FEM) and
2. the random effects model (REM), or error components model (ECM)

The Hausman Specification test can be used to decide between FEM and ECM. We can also use the Breusch–Pagan test to see if ECM is appropriate. Despite their substantial advantages, panel data pose several estimation and inference problems.

Hence the answer is **(D)** specification test.



Additional Note

Hausman Test: tests for the statistical significance of the difference between the coefficient estimates obtained by FE and by RE, under the null hypothesis that

H_0 : The RE estimates are efficient and consistent, and FE estimates are inefficient.

The test has a Wald test form, and is usually reported in χ^2 form with $k-1$ degrees of freedom (k is the number of regressors).

If $W < \text{critical value}$ then random effects is the preferred estimator.



Question 62

What is the exchange rate policy of Ethiopia

- a. Managed floating system
- b. Floating exchange system
- c. Fixed exchange rate system
- d. Pegged exchange rate system

Question Description	
Theme	Basic Macroeconomics
Course Name	Monetary Economics
Chapter	Chapter 6: Monetary policy in Ethiopia

Solution

The fixed exchange rate is the rate which is officially fixed (or pegged) in terms of gold or any other currency by the government and adjusted only infrequently.

The floating exchange rate is the rate which is determined by forces of supply and demand in the foreign exchange market. There is no (official) government intervention.

Managed floating is a hybrid of the fixed and floating exchange rate systems. It is characterized by some intervention in the exchange rate movements by the government of the country when a particular situation requires it. Currently, the exchange rate of the birr is determined by the managed floating exchange rate system.

Hence the answer is **(A)** Managed Floating System



Question 63

What is the particular underlying assumption behind the application of Microeconomic Household Theory of Fertility in low-income countries as distinct from developed countries?

- a. Other factors being held constant, the desired number of children vary inversely with household income, Education and Health
- b. Children are considered as a special kind of good outside the basket of consumption goods
- c. Children are considered as a special kind of investment good
- d. Fertility becomes independent area of decision outside of the rational economic response.

Question Description	
Theme	Basic Macroeconomics
Course Name	Development economics II
Chapter	Chapter 1: Population Growth and Economic Development

Solution

In the application of The Microeconomic Household Theory of Fertility, children are considered as a special kind of consumption (and in developing countries, particularly low-income countries, investment) good so that fertility becomes a rational economic response to the consumer's (family's) demand for children relative to other goods. The usual income and substitution effects are assumed to apply. That is, if other factors are held constant, the desired number of children can be expected to vary directly with household income (this direct relationship may not hold for poor societies; it depends on the strength of demand for children relative to other consumer goods and to the sources of increased income, such as female employment), inversely with the price (cost) of children, and inversely with the strength of tastes for other goods relative to children.

Hence the answer is **(C)** Children are considered as a special kind of investment good.



Question 64

What is the shortcoming of the new growth theory?

- a. it remains independent on a number of traditional neoclassical assumptions that are often adequate for LDCs
- b. it solves the issues of diminishing marginal product of capital
- c. it remains dependent on a number of traditional neoclassical assumptions that are often inadequate for LDCs
- d. it endogenizes technology

Question Description	
Theme	Basic Macroeconomics
Course Name	Development economics I
Chapter	Chapter 3: Growth Models and Theories of development

Solution

Endogenous growth theory (new growth theory): Economic growth generated by factors within the production process (e.g., increasing returns or induced technological change) that are studied as part of a growth model.

An important shortcoming of the new growth theory is that it remains dependent on a number of traditional neoclassical assumptions that are often inappropriate for developing economies. For example, it assumes that there is but a single sector of production or that all sectors are symmetrical. This does not permit the crucial growth-generating reallocation of labor and capital among the sectors that are transformed during the process of structural change. Moreover, economic growth in developing countries is frequently impeded by inefficiencies arising from poor infrastructure, inadequate institutional structures, and imperfect capital and goods markets. Because endogenous growth theory overlooks these very influential factors, its applicability for the study of economic development is limited, especially when country-to-country comparisons are involved.

Hence the answer is **(C)** it remains dependent on a number of traditional neoclassical assumptions that are often inappropriate for LDCs.



Question 65

What type of data can be used to study income of 2000 Ethiopian Rural households from 2000 to 2022?

- a. panel data
- b. Time series data
- c. Correctional
- d. Pooled panel data

Question Description	
Theme	Quantitative Methods in Economics with application
Course Name	Econometrics I
Chapter	Chapter 1: Introduction

Solution

Types of data

Time Series

- A time series is a set of observations on the values that a variable takes at different times.

Cross-section data

- Cross-section data are data on one or more variables collected at the same point in time.

Pooled Data (Pooled Cross sections)

- In pooled, or combined, data are elements of both time series and cross-section data.

Panel, Longitudinal, or Micro panel Data

- This is a special type of pooled data in which the same cross-sectional unit (say, a family or a firm) is surveyed over time.

Hence the answer is (A) panel data.



Note

It is very important to have a clear understanding of the four data types described above. I have provided an example for each type below.

<i>obsno</i>	<i>wage</i>	<i>educ</i>	<i>exper</i>	<i>female</i>	<i>married</i>
1	3.10	11	2	1	0
2	3.24	12	22	1	1
3	3.00	11	2	0	0
4	6.00	8	44	0	1
5	5.30	12	7	0	1
.	.	.	.	.	.
.	.	.	.	.	.
.	.	.	.	.	.
525	11.56	16	5	0	1
526	3.50	14	5	1	0

A Cross-Sectional Data Set on Wages and Other Individual Characteristics

<i>obsno</i>	<i>year</i>	<i>avgmin</i>	<i>avgcov</i>	<i>unemp</i>	<i>gnp</i>
1	1950	0.20	20.1	15.4	878.7
2	1951	0.21	20.7	16.0	925.0
3	1952	0.23	22.6	14.8	1015.9
.	.	.	.	.	.
.	.	.	.	.	.
.	.	.	.	.	.
37	1986	3.35	58.1	18.9	4281.6
38	1987	3.35	58.2	16.8	4496.7

A time series data on Minimum Wage, Unemployment, and Related Data for a certain country



<i>obsno</i>	<i>year</i>	<i>hprice</i>	<i>proptax</i>	<i>sqrft</i>	<i>bdrms</i>	<i>bthrms</i>
1	1993	85500	42	1600	3	2.0
2	1993	67300	36	1440	3	2.5
3	1993	134000	38	2000	4	2.5
.	.	.	.	.	.	.
.	.	.	.	.	.	.
.	.	.	.	.	.	.
250	1993	243600	41	2600	4	3.0
251	1995	65000	16	1250	2	1.0
252	1995	182400	20	2200	4	2.0
253	1995	97500	15	1540	3	2.0
.	.	.	.	.	.	.
.	.	.	.	.	.	.
.	.	.	.	.	.	.
520	1995	57200	16	1100	2	1.5

Pooled Cross Sections: Two Years of Housing Prices



<i>obsno</i>	<i>city</i>	<i>year</i>	<i>murders</i>	<i>population</i>	<i>unem</i>	<i>police</i>
1	1	1986	5	350000	8.7	440
2	1	1990	8	359200	7.2	471
3	2	1986	2	64300	5.4	75
4	2	1990	1	65100	5.5	75
.	.	.	.	.	.	.
.	.	.	.	.	.	.
.	.	.	.	.	.	.
297	149	1986	10	260700	9.6	286
298	149	1990	6	245000	9.8	334
299	150	1986	25	543000	4.3	520
300	150	1990	32	546200	5.2	493

Two-year panel data set



Question 66

What will happen to the slope of the aggregate demand curve, if money demand is more sensitive to a given change in interest rate?

- a. The AD curve become steeper reflecting output is not as such sensitive to change in the price level
- b. The AD curve become flatter reflecting output is as such sensitive to change in the price level
- c. The AD curve become steeper reflecting output is as such sensitive to change in the price level
- d. The higher the amount of shift in the LM curve due to changes in the monetary policy.

Question Description	
Theme	Basic Macroeconomics
Course Name	Macroeconomics I
Chapter	Chapter 3: Aggregated Demand in the closed economy

Solution

Changes in price level affect the real money supply and thereby the rate of interest. LM shifts leftward, and the rate of interest increases when the price level increases and the real money supply declines;

- higher interest rates have a smaller effect on equilibrium output when investment spending is less sensitive to interest rates.
- higher interest rates have larger effect on equilibrium output when investment spending is more sensitive to interest rates.

The interest sensitivity of demand for money also affects the change in output for a given change in price level. A certain aggregate demand schedule will be more steeply sloped than another aggregate demand schedule if the demand for money is more interest sensitive in the former demand schedule.

Hence the answer is **(C)** The AD curve become steeper reflecting output is as such sensitive to change in price level.



Question 67

What you call the following type model of non-stationary process? $y_t = \mu + y_{t-1} + \varepsilon_t$
28.282

- a. Random walk model with drift
- b. random walk model with constant variance
- c. random Walk model with constant auto covariance
- d. Random walk model trend

Question Description	
Theme	Quantitative Methods in Economics with application
Course Name	Econometrics II
Chapter	Chapter 3: Introduction to Basic Regression Analysis with Time Series Data

Solution

In Econometrics, although our interest is in stationary time series, one often encounters nonstationary time series, the classic example being the random walk model (RWM).

Consider the following models where u_t is a white noise error term and t is time measured chronologically.

Pure Random Walk (Random walk without drift)	$Y_t = Y_{t-1} + u_t$ $\Delta Y_t = Y_t - Y_{t-1} = u_t$ ▪ Constant mean ▪ Changing variance	REM, with or without a drift, ▪ Y_t will exhibit a stochastic trend in both cases. (Nonstationary stochastic processes) ▪ Both are difference stationary process (DSP). ▪ Both are characterized by persistence of random shocks (Infinite memory)
Random walk with drift	$Y_t = \beta_1 + Y_{t-1} + u_t$ $\Delta Y_t = Y_t - Y_{t-1} = \beta_1 + u_t$ β_1 is known as drift parameter. (Usually denoted by δ) ▪ Changing mean ▪ Changing variance	
Deterministic trend	$Y_t = \beta_1 + \beta_2 t + u_t$	It is a trend stationary process (TSP)

Hence the answer is (A) Random walk model with drift.



Question 68

When a country experiences a Trade Surplus

- a. Net Capital Outflow > 0
- b. Net Capital Outflow $= 0$
- c. Net Capital Outflow < 0
- d. Saving $<$ Investment

Question Description	
Theme	Basic Macroeconomics
Course Name	Macroeconomics II
Chapter	Chapter 4: Aggregate demand in an open Economy

Solution

$$S - I = NX \Rightarrow (\text{trade balance} = \text{net capital outflow})$$

This identity shows that an economy's net exports must always equal the difference between its saving and its investment.

$S - I$ is termed as net capital outflow/net foreign investment (domestic saving less domestic investment).

- If $S - I$ and NX are positive, we have a trade surplus. In this case, we are net lenders in world financial markets, and we are exporting more goods than we are importing.
- If $S - I$ and NX are negative, we have a trade deficit. In this case, we are net borrowers in world financial markets, and we are importing more goods than we are exporting.
- If $S - I$ and NX are exactly zero, we are said to have balanced trade because the value of imports equals the value of exports.

Hence the answer is **(A)** Net Capital Outflow > 0 .



Question 69

When a country's currency depreciates

- Foreigners find that its exports are cheaper and domestic residents find that imports from abroad are more expensive
- Foreigners find that its exports are cheaper and domestic residents find that imports from abroad are also cheaper.
- Domestic residents find that its exports are cheaper and foreigners find that imports from abroad are also cheaper
- Foreigners find that its exports are more expensive and domestic residents find that imports from abroad are cheaper

Question Description	
Theme	Basic Macroeconomics
Course Name	International Economics II
Chapter	Chapter 3: Balance of Payments

Solution

The balance of payments account of a country largely indicates the monetary aspect of its foreign trade, i.e., exports and imports. Also, exports earn foreign currency for a country, whereas imports imply the spending of foreign currency by the country. An increase in the value of the currency of a country makes its imports cheaper and its exports costlier (for foreign countries). In such a situation, imports increase and exports decrease, thus leading to a trade deficit, i.e., an unfavorable balance of payments. On the other hand, a decrease in the value of the currency of a country in terms of foreign currency, makes its imports costlier (for the local buyers) and exports cheaper (for the foreign buyers). In this situation, imports decrease and exports increase, thus leading to a trade surplus or favorable balance of payments.

Thus, for example, under a fixed exchange rate system

- if countries face a deficit in trade balance, they devalue their currency.
- if countries have a surplus in their trade balance, they revalue their currency.

Hence the answer is **(C)** Domestic residents find that its exports are cheaper and foreigners find that imports from abroad are also cheaper.



Question 70

When comparing alternative projects?

- a. Projects whose NPVR greater than the discount rate should be selected
- b. Net present value ratio (NPVR) is used
- c. Specific discounting rate and period for each project
- d. NPVR is computed by the ratio of the present value of investment to the NPV.

Solution

- The most widely used and straightforward discounted measure of project worth is the net present worth or the net present value (NPV).
- Having set the discount rate, an investment project is deemed acceptable if the discounted net benefits (benefits minus costs) is positive or non-negative.
- If there are alternative projects, how do you select from the available alternative projects? The NPV does not indicate the rate of return, in the sense that it does not directly indicate the magnitude of investment that generates the NPV.
- If one of several project alternatives has to be chosen, the project with the largest NPV should be selected. But we should know how much investment would be required to generate these positive NPVs if there are two or more alternatives. The ratio of the NPV and the present value of investment (PVI) should be considered and we get the net present value ratio (NPVR) when comparing alternative projects.

$$\text{NPVR} = \frac{\text{NPV}}{\text{PVI}}$$

- Given alternative projects, the one with the highest NPVR should be chosen. When comparing alternative projects, care should be taken to use the same discounting period and rate of discount rate for all projects.

Hence the answer is (B) Net present value ratio (NPVR) is used.



Question 71

When marginal revenue (MR) is negative,_____

- a. Price and the firm's total revenue moves in the same direction
- b. Price doesn't have any impact on the firm's total revenue
- c. The price elasticity of demand greater than one
- d. the price elasticity of demand is less than one

Question Description	
Theme	Basic Microeconomics
Course Name	Microeconomics I
Chapter	Chapter 3: The Theory of Production

Solution

This question can be approached using the same formula that we used in question 9.

$$MR = P - P \left(\frac{1}{e_d} \right)$$

- If elasticity of demand(e_d) is greater than one, MR is positive.
- If elasticity of demand(e_d) is less than one, MR is negative.

Hence the answer is **(D)** the price elasticity of demand is less than one.



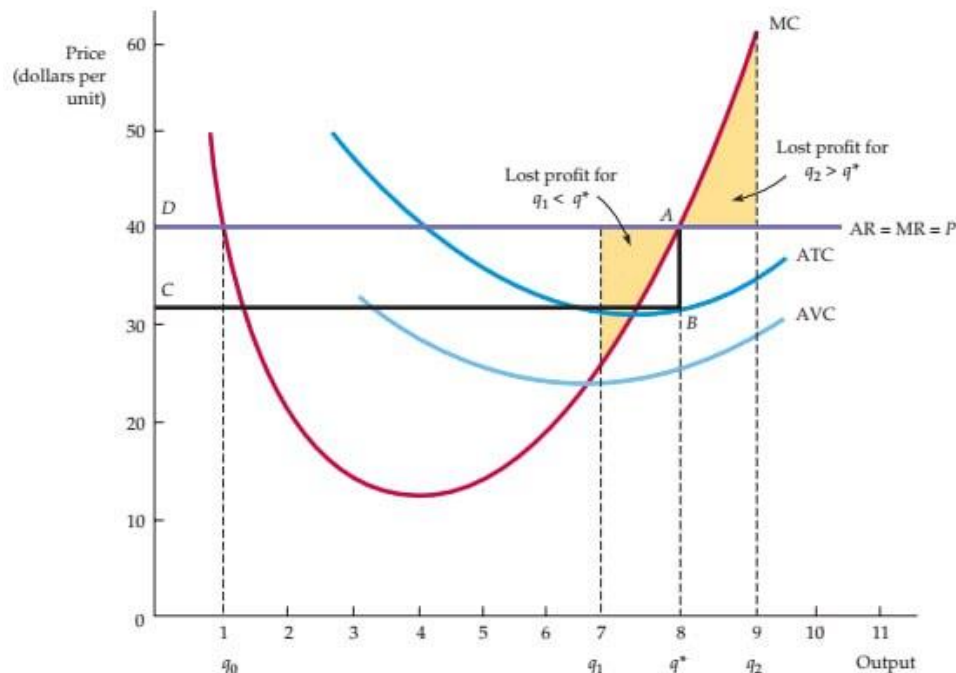
Question 72

When the competitive firm produces less than the equilibrium output

- The firm does not want to change the quantity that it sells
- The market price would be above its marginal cost
- The firm could increase its profit by reducing output
- The marginal cost would be above its price

Question Description	
Theme	Basic Microeconomics
Course Name	Microeconomics I
Chapter	Chapter 5: Perfect Competition Market

Solution



As can be seen from the graph of a perfectly competitive firm, If the firm produces less than the equilibrium output:

- The firm could increase profit by increasing output.
- The market price is above the marginal cost.

Hence the answer is **(B)** the market price would be above the marginal cost.



Question 73

Which of the following cannot be remedy to remove perfect multicollinearity from the model?

- a. Transforming the functional relationship
- b. Increasing the size of the sample data
- c. Decreasing the size of the sample data
- d. Dropping one of the highly collinear variables

Question Description	
Theme	Quantitative Methods in Economics with application
Course Name	Econometrics I
Chapter	Chapter 4: Violations of the Assumptions of the Classical Linear Regression Model.

Solution

Multicollinearity refers to a linear relationship among some or all explanatory variables of a multiple linear regression model.

One can try the following rules of thumb to address the problem of multicollinearity.

1. A priori information
2. Combining cross-sectional and time series data
3. Dropping a variable(s) and specification bias
4. Transformation of variables
5. Additional (new) data
6. Reducing collinearity in polynomial regressions

Hence the answer is (C) decreasing the size of the sample data.



Question 74

Which of the following changes shifts the long-run aggregate supply curve to the right?

- a. A decrease in taxes.
- b. An increase in labor supply
- c. A decrease in the demand for labor
- d. An increase in consumer confidence

Question Description	
Theme	Basic Macroeconomics
Course Name	Macroeconomics I
Chapter	Chapter 5: Aggregate Supply

Solution

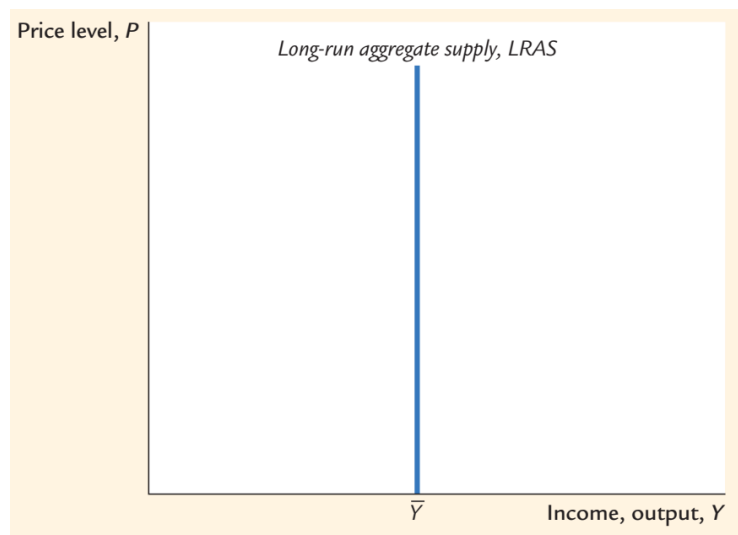


Figure: The long run aggregate supply curve

In long run, the Aggregate supply is not affected by the changes in prices. But rather the various factors that can affect the aggregate supply are the changes in labor supply, technological changes etc.

When a demographic change increases the labor supply, the productivity increases in the long run. This will cause the aggregate supply curve in the long run to shift to the right.

Hence the answer is **(B)** an increase in labor supply.

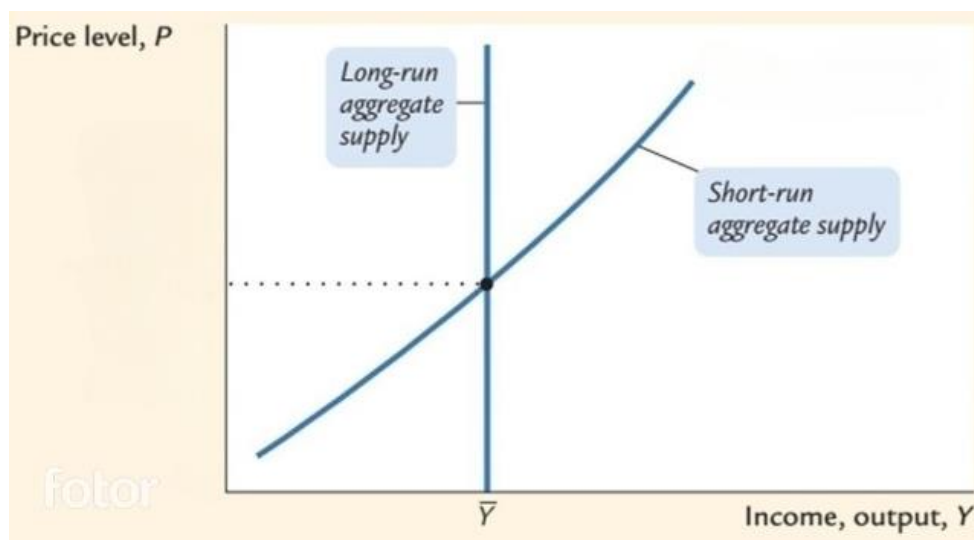


Additional Note

Because the firms that supply goods and services have flexible prices in the long run but sticky prices in the short run, the aggregate supply relationship depends on the time horizon. Accordingly, there are two different aggregate supply curves: the long-run aggregate supply curve LRAS and the short-run aggregate supply curve SRAS.

The Long Run: The Vertical Aggregate Supply Curve

According to the classical model, output the amount of output produced depends on the fixed amounts of capital and labor and on the available technology and not on the price level. To show that output is fixed regardless of the price level, we draw a vertical aggregate supply curve, as shown below. This long run level of output, $\bar{Y}$, is called the full-employment, or natural, level of output.



The Short Run: The Non-Vertical Aggregate Supply Curve

In the short run, some prices are sticky and therefore do not adjust to changes in demand. Because of this price stickiness, the short-run aggregate supply curve is not vertical. Market imperfection causes the output of the economy to deviate from its natural level. As a result, the short-run aggregate supply curve is upward sloping rather than vertical as shown above.



Question 75

Which of the following conditions should be fulfilled to consider a good as tradable?

- a. If its external demand or external supply could be regarded as fixed
- b. F.O.B price is less than domestic cost of production(DCP)
- c. If exported through government intervention by use of export subsidies
- d. DCP is less than C.I.F price

Question Description	
Theme	Basic Microeconomics
Course Name	Development Planning and Project Analysis II
Chapter	Chapter 3: Economic Analysis of Projects

Solution

- To determine whether a good is tradable or non-tradable, the first step is to ascertain whether the good trades internationally. If no international trade exists, then it is safe to assume that the good is non tradable.
- In order to adjust financial prices to economic values, we need to do an adjustment for distortions in market prices of traded items.
- Tradable (Traded items) are those for which, if exports,

$\text{f.o.b. price} > \text{domestic cost of production},$

or the items may be exported through government intervention by use of export subsidies and the like, and, if imports,

$\text{domestic cost of production} > \text{c.i.f. price}.$

Hence the answer is **(C)** if exported through government intervention by use of export subsidies.

Additional Note

Nontraded (Non-tradable) items are those for which

$\text{c.i.f. price} > \text{domestic cost of production} > \text{f.o.b. price},$

or the items are nontraded because of government intervention by means of import bans, quotas, and the like.



Question 76

Which of the following is an example indirect finance in Ethiopian context?

- a. A farmer borrowing money from his nearby shopkeeper
- b. A farmer borrowing money from local money lender
- c. A farmer borrowing money from the Commercial Bank of Ethiopia
- d. A farmer borrowing money from his relatives.

Question Description	
Theme	Basic Macroeconomics
Course Name	Monetary Economics
Chapter	Chapter 2: An overview of the financial system

Solution

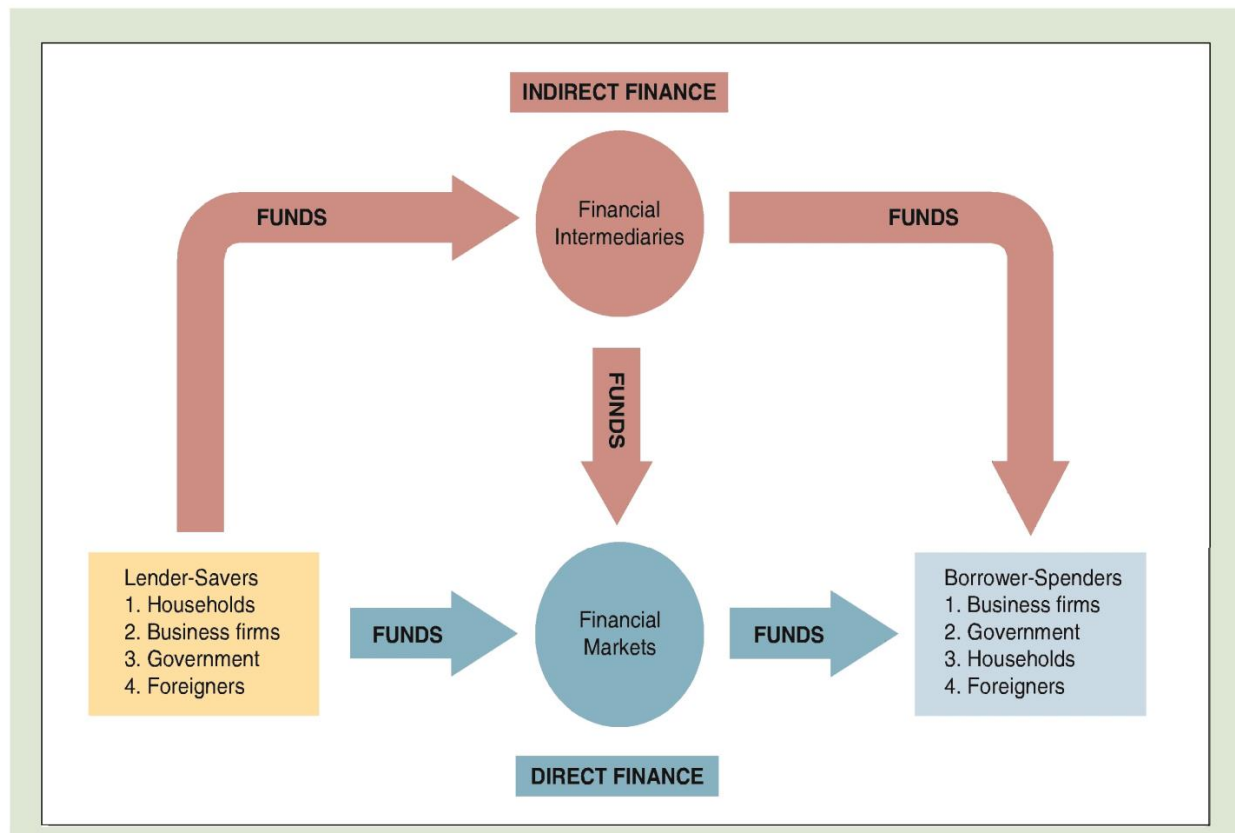


Figure: Flow of funds through the Financial System

Hence the answer is **(C)** A farmer borrowing money from Commercial Bank of Ethiopia.



Question 77

Which of the following is implied by the comparative advantage theory of trade regarding nations that produce two goods?

- a. If it is impossible for one nation to have a comparative advantage in the production of both goods it should follow its absolute cost advantage
- b. It is impossible for one nation to have a comparative advantage in the production of both goods.
- c. If it is impossible for one nation to have a comparative advantage in the production of both goods it should abandon all
- d. It is possible for every nation to produce two goods and enjoy comparative advantage from both

Question Description	
Theme	Basic Macroeconomics
Course Name	International Economics I
Chapter	Chapter 2: International Trade Theories

Solution

The Law of Comparative Advantage

According to the law of comparative advantage, even if one nation is less efficient than (has an absolute disadvantage with respect to) the other nation in the production of both commodities, there is still a basis for mutually beneficial trade. The first nation should specialize in the production and export of the commodity in which its absolute disadvantage is smaller (this is the commodity of its comparative advantage) and import the commodity in which its absolute disadvantage is greater (this is the commodity of its comparative disadvantage).

Note that in a two-nation, two-commodity world, once it is determined that one nation has a comparative advantage in one commodity, then the other nation must necessarily have a comparative advantage in the other commodity.

Hence the answer is **(B)** it is impossible for one nation to have a comparative advantage in the production of both goods.



Additional Note

Absolute Advantage

According to Adam Smith, trade between two nations is based on absolute advantage. When one nation is more efficient than (or has an absolute advantage over) another in the production of one commodity but is less efficient than (or has an absolute disadvantage with respect to) the other nation in producing a second commodity, then both nations can gain by each specializing in the production of the commodity of its absolute advantage and exchanging part of its output with the other nation for the commodity of its absolute disadvantage.



Question 78

Which of the following is the result of a policy that reduces national saving such as budget deficit?

- a. Reduces the supply of loanable funds and drives up the interest rate. The dollar appreciates, and net exports fall
- b. Reduces the supply of loanable funds and drives down the interest rate. The dollar appreciates, and net exports fall
- c. Reduces the supply of loanable funds and drives up the interest rate. The dollar depreciates, and net exports fall
- d. Reduces the supply of loanable funds and drives up the interest rate. The dollar appreciates, and net exports rise

Solution

In an open economy, government budget deficits reduce the supply of loanable funds, drive up the interest rate, crowd out domestic investment and cause net foreign investment to fall.

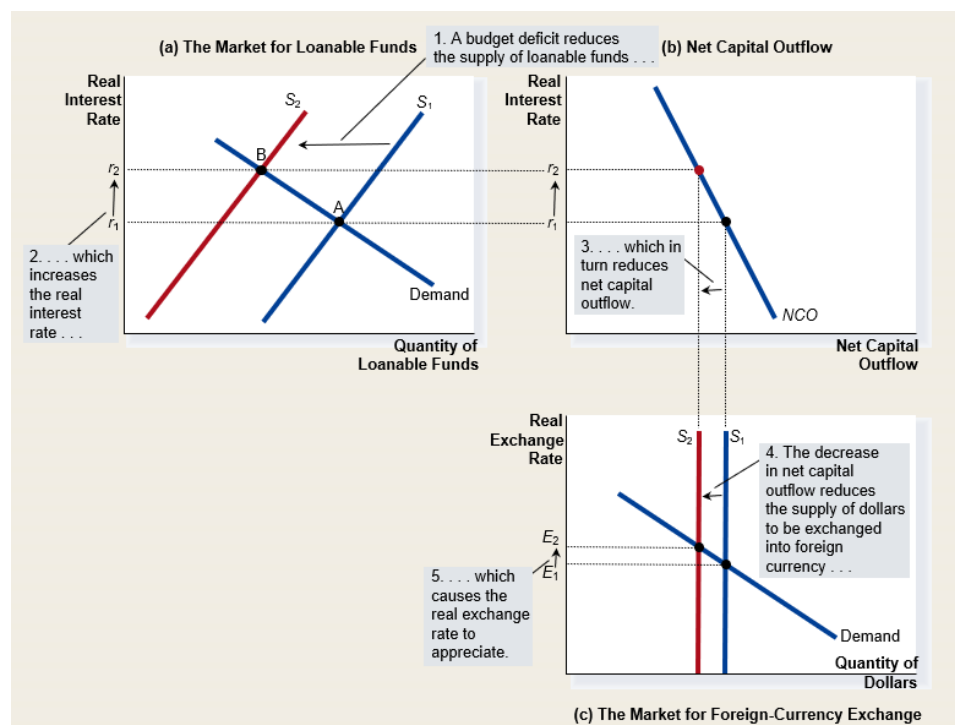


Figure: The effects of government budget deficit

Hence the answer is **(A)** reduces the supply of loanable funds and drives up the interest rate. The dollar appreciates, and net exports fall.



Question 79

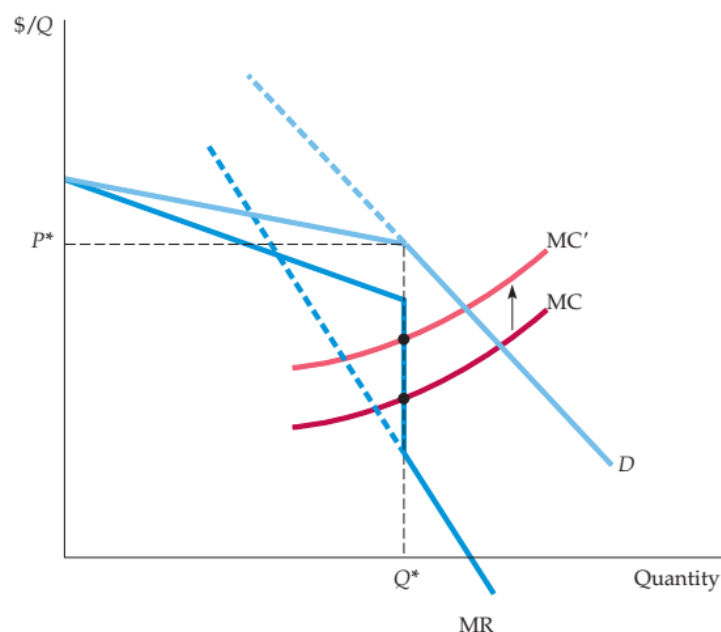
Which of the following is true about the kinked demand curve model?

- a. The upper part of the demand curve have low price elasticity while the lower part has high price elasticity
- b. A planned demand curve and the market share curve are combined to form kinked demand curve
- c. If the cost of firms goes down and if the demand for their products falls, they may prefer to lower their prices.
- d. If the cost of firms goes up and demand rises, they may prefer to raise their prices.

Question Description	
Theme	Basic Microeconomics
Course Name	Microeconomics II
Chapter	Chapter 2: Oligopoly

Solution

Kinked demand curve is model oligopoly model in which each firm faces a demand curve kinked at the currently prevailing price: at higher prices demand is very elastic, whereas at lower prices it is inelastic.



Each firm believes that if it raises its price above the current price P^* , none of its competitors will follow suit, so it will lose most of its sales. Each firm also believes that if it lowers price, everyone will follow suit, and its sales will increase only to the extent that market demand increases. As a result, the firm's demand curve D is kinked at price P^* , and its marginal revenue curve MR is discontinuous at that point. If

marginal cost increases from MC to MC' , the firm will still produce the same output level Q^* and charge the same price P^* .



Question 80

Which one of the following best explains the Fisher's theory of consumption?

- a. The timing of the income is relevant because consumers can lend but cannot borrow
- b. The timing of the income is irrelevant because consumers can lend and borrow
- c. The timing of the income is irrelevant because consumers cannot lend and borrow
- d. The timing of the income is relevant because consumers cannot lend and borrow

Question Description	
Theme	Basic Macroeconomics
Course Name	Macroeconomics II
Chapter	Chapter 1: Behavioral Foundations: Theories of Consumption

Solution

The economist Irving Fisher developed the model with which economists analyze how rational, forward-looking consumers make intertemporal choices—that is, choices involving different periods of time. Fisher's model illuminates the constraints consumers face, the preferences they have, and how these constraints and preferences together determine their choices about consumption and saving

Keynes Theory of Consumption: Current consumption depends only on current income.

Fisher's Theory of Consumption: Current consumption depends only on the present value of lifetime income. The timing of income is irrelevant because the consumer can borrow or lend between periods.

Hence the answer is **(B)** the timing of the income is irrelevant because consumers can lend and borrow.



Question 81

Which one of the following best relates WTO and the Uruguay Rounds?

- a. WTO monitors each member country's trade policies with respect to the trade agreements that were made in the Uruguay Round
- b. WTO agreements do not monitor promises to reduce barriers to trade that the Uruguay Round attended
- c. WTO monitors trade policies on non-members with respect to the trade agreements that were made in the Uruguay Round
- d. The goal of the WTO is different from the goal of the Uruguay round with respect to promoting trade liberalization and fostering economic growth and development.

Question Description	
Theme	Basic Macroeconomics
Course Name	International Economics I
Chapter	Chapter 5: International Trade Institutions and Regional Economic Arrangements

Solution

In December 1993, the Uruguay Round, the eighth and most ambitious round of multilateral trade negotiations in history, in which 123 countries participated, was completed after seven years of tortuous negotiations.

The major provisions of the accord were the following.

1. Tariffs. Tariffs on industrial products were to be reduced from an average of 4.7 percent to 3 percent, and the share of goods with zero tariffs was to increase from 20–22 percent to 40–45 percent; tariffs were removed altogether on pharmaceuticals, construction equipment, medical equipment, paper products, and steel.
2. Quotas. Nations were to replace quotas on agricultural imports and imports of textiles and apparel (under the Multifiber Agreement) with less restrictive tariffs by the end of 1999 for agricultural products and by the end of 2004 for textiles and apparel; tariffs on agricultural products were to be reduced by 24 percent in developing nations and by 36 percent in industrial nations, and tariffs on textiles were to be cut by 25 percent.



3. Antidumping. The agreement provided for tougher and quicker action to resolve disputes resulting from the use of antidumping laws, but it did not ban their use.
4. Subsidies. The volume of subsidized agricultural exports was to be reduced by 21 percent over a six-year period; government subsidies for industrial research were limited to 50 percent of applied research costs.
5. Safeguards. Nations could temporarily raise tariffs or other restrictions against an import surge that severely harmed domestic industry, but it barred countries from administering health and safety standards unless based on scientific evidence and not simply to restrict trade.
6. Intellectual property. The agreement provided for 20-year protection of patents, trademarks, and copyrights, but it allowed a 10-year phase-in period for patent protection in pharmaceuticals for developing countries.
7. Services. The United States failed to secure access to the markets of Japan, Korea, and many developing nations for its banks and security firms, and did not succeed in having France and the European Union lift restrictions on the showing of American films and TV programs in Europe.
8. Other industry provisions. The United States and Europe agreed to continue talking about further limiting government subsidies to civil aircraft makers, opening up the distance telephone market, and limiting European subsidies to steelmakers; the United States also indicated that it intended to continue negotiating the further opening of the Japanese computer chip market.
9. Trade-related investment measures. The agreement phased out the requirement that foreign investors (such as automakers) buy supplies locally or export as much as they import.
10. World Trade Organization. The agreement also called for the replacement of the General Agreement on Tariffs and Trade (GATT) secretariat with the World Trade Organization (WTO) in Geneva with authority not only in trade in industrial products but also in agricultural products and services. Trade disputes were also to be settled by a vote of two-thirds or three-quarters of the nations rather than unanimously as under GATT (which meant that the guilty nation could block any action against it).

Hence the answer is (A) WTO monitors each member country's trade policies with respect to the trade agreements that were made in Uruguay Round.



Question 82

Which one of the following better depicts the effects of import quotas and import tariffs on the level of imports?

- a. An import quota limits imports to the specified level with certainty, while the trade effect of an import tariff may be uncertain.
- b. In both cases it is uncertain whether imports will be limited to the specified level
- c. An import quota limits imports to the specified level is uncertain, while the trade effect of an import tariff is certain.
- d. An import quota limits imports less than the level import tariff does.

Question Description	
Theme	Basic Macroeconomics
Course Name	International Economics I
Chapter	Chapter 3: International Trade Theories

Solution

As a measure towards protection of domestic industries of a country, sometimes restrictions are imposed on foreign trade, particularly on imports. These restrictions are broadly of two types.

- **Tariff Restrictions:** Tariff restrictions are in the form of taxes on the import of goods, called custom duty or import duty. Such taxes raise the price of imported goods in the domestic market. These high prices of imported goods are expected to reduce their demand in the domestic market and thus to act to restrict imports. Tariff increases government revenue.
- **Quantitative Restriction (Import Quota):** These restrictions take the shape of fixing the maximum quantity of goods that is permitted to be imported. Thus, the government may determine the total import quota of goods, i.e., the total amount of goods that can be imported, and can allot this quota to various importers. Nothing beyond the quota is allowed to be imported. This naturally limits the quantity of imports.

Hence the answer is (A) an import quota limits imports to the specified level with certainty, while the trade effect of an import tariff may be uncertain.



Question 83

Which one of the following conceptions of development takes development as a multidimensional process rather than growth?

- a. Development mainly involves the acceleration of economic growth, that trickles down to the fulfilment of basic material needs
- b. Development is a movement away from unsatisfactory life toward a situation or condition of life regarded as materially and spiritually better.
- c. Development involves minor changes in social structures, popular attitudes, and national institutions
- d. Development sometimes involves accelerated economic growth, with reduction of inequality, and the eradication of poverty.

Question Description	
Theme	Basic Macroeconomics
Course Name	Development economics I
Chapter	Chapter 1: Economics of Development: Concepts and Approaches

Solution

Development must be conceived of as a multidimensional process involving major changes in social structures, popular attitudes, and national institutions, as well as the acceleration of economic growth, the reduction of inequality, and the eradication of poverty. Development, in its essence, must represent the whole gamut of change by which an entire social system, tuned to the diverse basic needs and evolving aspirations of individuals and social groups within that system, moves away from a condition of life widely perceived as unsatisfactory toward a situation or condition of life regarded as materially and spiritually better. No one has identified the human goals of economic development as well as Amartya Sen, perhaps the leading thinker on the meaning of development.

Hence the answer is **(B)** Development is a movement away from unsatisfactory life toward a situation or condition of life regarded as materially and spiritually better.



Question 84

Which one of the following describes the unintended consequences of import substitution strategy by blocking the hoped-for process of self-reliant integrated industrialization?

- a. Purchasing their own inputs from overseas sources of supply rather than through backward linkages to domestic suppliers
- b. Using imported consumer goods other than they produce to the detriment of domestic companies
- c. Decreasing the costs of inputs to potentially forward-linked industries
- d. Selling to foreign markets while domestic markets are starved.

Question Description	
Theme	Basic Macroeconomics
Course Name	Development economics II
Chapter	Chapter 6: Foreign aid, debt, financial reform and development

Solution

Most observers agree that the import-substituting (IS) strategy of industrialization has been largely unsuccessful. Specifically, there have been five undesirable outcomes.

1. Secure behind protective tariff walls and immune from competitive pressures, many IS industries (both publicly and privately owned) remain inefficient and costly to operate.
2. The main beneficiaries of the import substitution process have been the foreign firms that were able to locate behind tariff walls and take advantage of liberal tax and investment incentives.
3. Most import substitution has been made possible by the heavy and often government-subsidized importation of capital goods and intermediate products by foreign and domestic companies.
4. A fourth detrimental effect of many import substitution strategies has been their impact on traditional primary-product exports.
5. Import substitution, which may have been conceived with the idea of stimulating infant-industry growth and self-sustained industrialization by creating “forward” and “backward” linkages with the rest of the economy, has often inhibited that industrialization. Many infants never grow up,



content to hide behind protective tariffs and governments loath to force them to be more competitive by lowering tariffs. In fact, governments themselves often operate protected industries as state-owned enterprises. Moreover, by increasing the costs of inputs to potentially forward-linked industries (those that purchase the output of the protected firm as inputs or intermediate products in their own productive process, such as a printer's purchase of paper from a locally protected paper mill) and by purchasing their own inputs from overseas sources of supply rather than through backward linkages to domestic suppliers, inefficient import-substituting firms may in fact block the hoped-for process of self-reliant integrated industrialization.

Hence the answer is (A) purchasing their own inputs from overseas sources of supply rather than through backward linkages to domestic suppliers.



Question 85

Which one of the following is a point of differences between the views of neoclassical counter revolution theory of development and dependence theorists of development?

- a. Neoclassical counterrevolutionaries argue that the developing world is underdeveloped, because of the heavy hand of the state and lack of economic incentives
- b. Neoclassical counterrevolutionaries argue that underdevelopment is not because of the heavy hand of the state and not because of the lack of economic incentives
- c. Dependence theorists argue that under development is not associated with the predatory activities of the developed world
- d. Neoclassical counterrevolution claim that the international economic system, has a heavy impact on dualistic developing economies,

Question Description	
Theme	Basic Macroeconomics
Course Name	Development economics I
Chapter	Chapter 3: Growth Models and Theories of development

Solution

Contrary to the claims of the dependence theorists, the neoclassical counterrevolutionaries argue that the developing world is underdeveloped not because of the predatory activities of the developed world and the international agencies that it controls but rather because of the heavy hand of the state and the corruption, inefficiency, and lack of economic incentives that permeate the economies of developing nations.

According to the neoclassical counterrevolutionaries, what is needed is not a reform of the international economic system, a restructuring of dualistic developing economies, an increase in foreign aid, attempts to control population growth, or a more effective development planning system. Rather, it is simply a matter of promoting free markets and laissez-faire economics within the context of permissive governments that allow the “magic of the marketplace” and the “invisible hand” of market prices to guide resource allocation and stimulate economic development.

Hence the answer is **(A)** neoclassical counterrevolutionaries argue the developing world is underdeveloped, because of the heavy hand of the state and lack of economic incentives.



Question 86

Which one of the following is a valid conclusion that may be made if the existence of environmental degradations, needs for R&D and extension services in agriculture, economies of scale in agricultural marketing, information asymmetries in agricultural product quality, missing markets, and monopoly power in input supply, are confirmed cases in agriculture?

- a. The confirmed cases of market failures are cases that do not create incentive problems and manageable by the market itself.
- b. In fact, market failures in the sector are quite common but they do not need government role to rectify them
- c. Although agriculture generally looks a perfectly competitive activity, it is full of market failures and is in need of active government role
- d. It can still be stated that there are no market failures and no role for government.

Question Description	
Theme	Basic Microeconomics
Course Name	Microeconomics II
Chapter	Chapter 6: Introduction to Asymmetric Information

Solution

Market failure refers to the inefficient allocation of resources that occurs when individuals acting in rational self-interest produce a less-than-optimal outcome.

It may be possible to correct market failures using private market solutions, government-imposed solutions, or voluntary collective actions.

Causes of Market Failure

1. **Externalities:** Externalities occur when the consumption of a good or service benefits or harms a third party. Pollution resulting from the production of certain goods is an example of a negative externality that can hurt individuals and communities. The collateral damage caused by negative externalities may lead to market failure.
2. **Information failure:** When there is insufficient information available to certain participants in the market, this can also be the source of market failure. If the buyer or seller in a transaction lacks access to the information on which the price is based, they may be willing to overpay or undercharge for a good or service, disrupting the market's equilibrium.



3. Market control: When one party has too much control over a market, this can also create imbalanced pricing and lead to market failure. In the case of a monopoly or oligopoly, a single seller or a small group of sellers can manipulate pricing. In other situations, known as monopsony or oligopsony, it is the buyers that have the advantage. In either case, the disrupted balance of supply and demand could cause market failure.
4. Public goods: Public goods are another example of market failure because they defy the tenets of supply and demand that drive the free markets. Public goods and services are nonexcludable—once something like a street light is produced, it is accessible to everyone, and the producer cannot limit consumption only to paying customers. Public goods are also nonrival, as use by one individual does not limit consumption by others. Given these characteristics, the private sector has little incentive to produce public goods, which leads to market failure, and the government usually has to provide these goods or subsidize their production.

Hence the answer is (C) Although agriculture generally looks a perfectly competitive activity, it is full of market failures and is in need of active government role.



Question 87

Which one of the following is not an objective of project evaluation?

- a. Determining the problems and challenges involved in the execution of a project
- b. Responding to the demands of customers.
- c. Identifying the progress of the project.
- d. Completing the project within the allotted time and resources.

Question Description	
Theme	Basic Microeconomics
Course Name	Development Planning and Project Analysis II
Chapter	Chapter 3: Economic Analysis of Projects

Solution

Evaluation: is the last phase of the project cycle. Evaluation is a comparative analysis between what the project was intended to become, in terms of timing and quantity of costs and benefits, and what actually has happened.

Ex-evaluations are means of deriving lessons on a given project's undertaking. The analyst looks systematically at the elements of success and failure indicators in the project experience to learn how better to plan for the future.

Evaluation is not limited only to completed projects. Evaluation may be undertaken anytime when the project is in trouble as a first step in a re-planning effort (projects can be terminated based on evaluation reports). Evaluation may be undertaken by project management, sponsoring agency, external evaluators, separate monitoring unit of the project and/or any appropriate government planning unit. In any case, the project plan should be reviewed to see if it was an appropriate one in light of the objects set forth. Each objective should be examined to determine whether it was considered carefully and whether appropriate provision was made. Evaluators look into the basic questions: Was the technology proposed appropriate? Were the institutional, organizational, and managerial arrangements suited to the conditions? Were the financial aspects carefully worked out on the basis of realistic assumptions? Were the economic implications properly explored? And how did the project in practice compare with each aspect of the project planning? The evaluation should consider the response of project management and the sponsoring agencies to changing circumstances. Did



management respond quickly enough to changes? Was its response carefully considered and appropriate? Did the institutional and organizational structure in the project permit a flexible response? How could a project's structure be altered to make the response to change more flexible and appropriate in the future? From the evaluation should come carefully considered recommendations about how to improve the appropriateness of each aspect of the project design so that plans for project implementation should be revised if the project ongoing.

Based on the evaluation report, a project has a chance of being rejected, accepted (implemented as it is), modified, or delayed for some years.

Hence the correct answer is (B) responding to the demands of customers.



Question 88

Which one of the following is not the effect of complementarities?

- a. Complementarities necessitate a policy that assists the economy to move from a suboptimal outcome to one that is more efficient.
- b. Complementarities create externalities that may precipitate a coordination failure
- c. Complementarities create case of several sectors simultaneously functioning at a suboptimal level
- d. Complementarity effects are free from history and expectations

Question Description	
Theme	Basic Macroeconomics
Course Name	Development economics I
Chapter	Chapter 4: Historic Growth and Contemporary Development: Lessons and controversies

Solution

Complementarity: An action taken by one firm, worker, or organization that increases the incentives for other agents to take similar actions. Complementarities often involve investments whose return depends on other investments being made by other agents.

Coordination failure: A state of affairs in which the inability of agents to coordinate their behavior (choices) leads to an outcome (equilibrium) that leaves all agents worse off than in an alternative situation that is also an equilibrium.

The coordination failures that may arise in the presence of complementarities highlight potential policies for deep interventions that move the economy to a preferred equilibrium or even to a higher permanent rate of growth that can then be self-sustaining. The other edge of the sword, however, is that with deep interventions, the potential costs of a public role become much larger. The results of corrupt privatization in countries have been devastating for its economy, preventing it from enjoying the benefits of the market and potentially keeping it in a suboptimal equilibrium for many years to come.

Hence the answer is **(D)** Complementarity effects are free from history and expectations.



Question 89

Which one of the following is the reason for the labour supply curve to have a segment with negative slope

- As incomes reach the level required for a comfortable standard of living, workers put forward greater demands for more leisure.
- As the wage rate increases, the income effect always dominates the substitution effect and hence enables the worker to have more leisure activities
- beyond a certain level of the wage rate, the substitution effect of a wage rate change dominates the income effect and hence the worker prefers to work more
- As the wage rate increases, the substitution effect always dominates the income effect and hence enables the worker to have more work activities

Question Description	
Theme	Basic Macroeconomics
Course Name	Macroeconomics II
Chapter	Chapter 4: The Labor Market

Solution

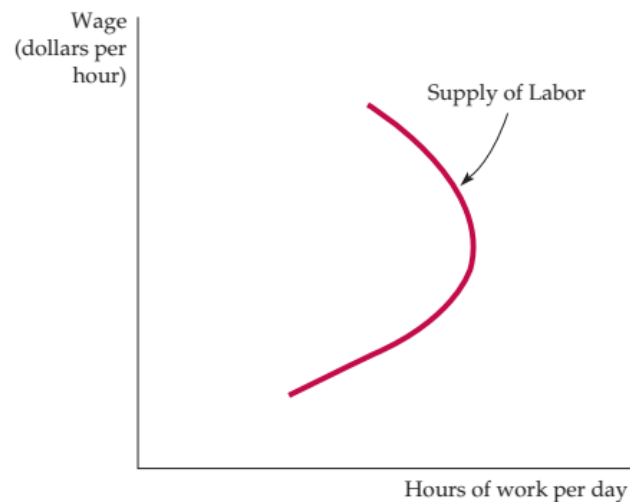


Figure: Backward bending supply of labour

When the wage rate increases, the hours of work supplied increase initially but can eventually decrease as individuals choose to enjoy more leisure and to work less. The backward-bending portion of the labor supply curve arises when the income effect of the higher wage (which encourages more leisure) is greater than the substitution effect (which encourages more work).



As the standard of living increases people find that unless they have the time to enjoy leisure activities it is not worth their while to work harder in order to obtain the higher income required for more leisure. Thus, as incomes reach the level required for a comfortable standard of living, workers put forward greater demands (in labour negotiations) for more holidays, longer vacations, shorter work weeks, fewer hours per working day rather than demanding ever higher wage rates associated with longer working hours.

Hence the answer is (A) As incomes reach the level required for a comfortable standard of living, workers put forward greater demands for more leisure.



Question 90

Which one of the following represents a policy of ad valorem tariff? Imposing tariff at

- a. Combination of a fixed percentage and a fixed sum of the value of the traded commodity
- b. Difference of a fixed percentage and a fixed sum of the value of the traded commodity
- c. Fixed sum per physical unit of the traded commodity.
- d. Fixed percentage of the value of the traded commodity.

Question Description	
Theme	Basic Macroeconomics
Course Name	International Economics I
Chapter	Chapter 3:

Solution

The most important type of trade restriction has historically been the tariff. A tariff is a tax or duty levied on the traded commodity as it crosses a national boundary.

Tariffs can be ad valorem, specific, or compound.

- The ad valorem tariff is expressed as a fixed percentage of the value of the traded commodity.
- The specific tariff is expressed as a fixed sum per physical unit of the traded commodity.
- A compound tariff is a combination of an ad valorem and a specific tariff.

For example, a 10 percent ad valorem tariff on bicycles would result in the payment to customs officials of the sum of \$10 on each \$100 imported bicycle and the sum of \$20 on each \$200 imported bicycle. On the other hand, a specific tariff of \$10 on imported bicycles means that customs officials collect the fixed sum of \$10 on each imported bicycle regardless of its price. Finally, a compound duty of 5 percent ad valorem and a specific duty of \$10 on imported bicycles would result in the collection by customs officials of the sum of \$15 on each \$100 bicycle and \$20 on each \$200 imported bicycle.

Hence the answer is **(D)** fixed percentage of the value of the traded commodity.



Question 91

Which one of the following sequential order in project cycle is correct?

- a. Pre- feasibility → Feasibility → Identification → Selection → Implementation → Ex-post evaluation
- b. Identification → Pre- feasibility → Feasibility → Selection → Implementation → Ex-post evaluation
- c. Identification → Pre- feasibility → Feasibility → Implementation → Selection → Ex-post evaluation
- d. Pre- feasibility → identification → Feasibility → Selection → Implementation → Ex-post evaluation

Question Description	
Theme	Basic Microeconomics
Course Name	Development Planning and Project Analysis II
Chapter	Chapter 1: Basic Concepts

Solution

Project cycle involves the following stages.

1. Project Identification (Opportunity Stage)
 - 1.1. Project Concept Definition Phase
 - 1.2. Project Identification (Investigation) Phase
2. Project Preparation and Analysis Stage
 - 2.1. Pre-Feasibility Study (Preliminary Screening) Phase
 - 2.2. Feasibility Study Phase
3. Project Appraisal Stage
4. Project Implementation Stage
 - 4.1. Pre-Investment Phase
 - 4.2. Investment Phase
 - 4.3. Operational Phase
 - 4.3.1. Commissioning and starting of commercial production
 - 4.3.2. Ex-post project evaluation
 - 4.3.3. Replacements/Rehabilitation
 - 4.3.4. Expansion/Innovation
5. Evaluation and Supervision

The best answer is **(B)** Identification → Pre-Feasibility → Feasibility → Selection → Implementation → Ex-post evaluation



Question 92

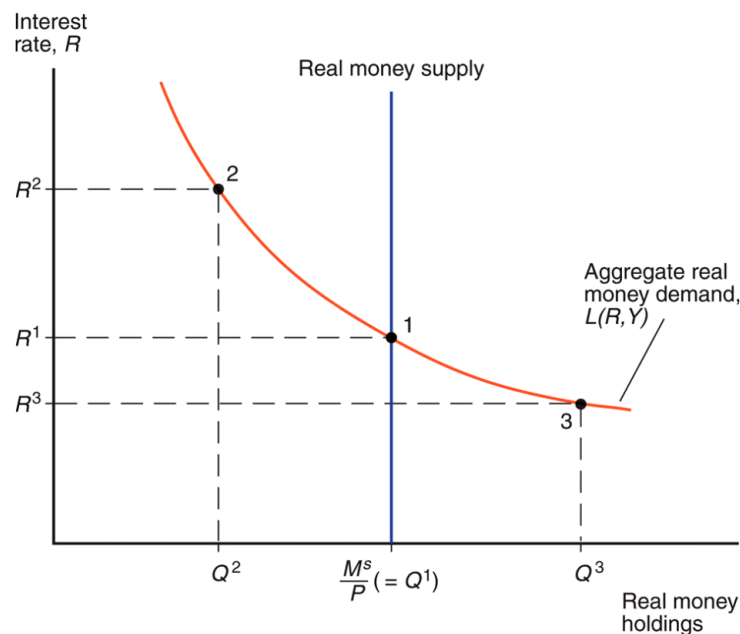
Which one of the following statements holds at equilibrium of the money market?

- a. Aggregate supply of money is equal to the total demand for money by the government
- b. Aggregate supply of money is equal to the aggregate demand for money.
- c. Aggregate demand supply of money is equal to the total demand for money by firms
- d. Aggregate supply of money is equal to the total demand for money by households

Question Description	
Theme	Basic Macroeconomics
Course Name	Macroeconomics I
Chapter	Chapter 3: Aggregate demand in a closed economy

Solution

The money market equilibrium can be depicted by the graph below.



With P and Y given and a real money supply of M^s/P , money market equilibrium is at point 1. At this point aggregate real money demand and the real money supply are equal and the equilibrium interest rate is R^1 .

Hence the answer is **(B)** aggregate supply of money is equal to the aggregate demand for money,



Question 93

Which one of the following time paths of profit $\Pi(t)$ can be derived from the differential equation relating profit Π , time rate of output $\frac{d\Pi}{dt}$ and time t as $\frac{d\Pi}{dt} = 4t$;

- a. $\Pi(t) = Ae^{-2t^2} + 1$
- b. $\Pi(t) = e^{-2t^2} + 10$
- c. $\Pi(t) = Ae^{2t^2} + 4$
- d. $\Pi(t) = Ae^{-2t} + 1$

Question Description	
Theme	Quantitative Methods in Economics with application
Course Name	Mathematical Economics
Chapter	Chapter 6: Dynamic Optimization

Solution

The equations were not legible on the exam. The latex equations embedded in the text has been converted to an image and rewritten as follows.

Which one of the following time paths of profit $\Pi(t)$ can be derived from the differential equation relating profit Π , time rate of output $\frac{d\Pi}{dt}$ and time t as $\frac{d\Pi}{dt} = 4t$;

Yet, there still seems to be a type error in this question. The proper differential equation the question intended to use isn't clear. But here are the formulas that could have been used to solve the question.

Note

For a homogeneous linear differential equation $\frac{dy}{dt} + ay = 0$, the general solution is given by $y(t) = Ae^{-at}$

For a nonhomogeneous linear differential equation $\frac{dy}{dt} + ay = b$, the general solution is given by $y(t) = Ae^{-at} + \frac{b}{a}$ ($a \neq 0$)



Question 94

With the value of human capital as an investment written as $\sum_{t=0}^T \frac{E_t - N_t}{(1+r)^t}$ where E is income with extra education, N is income without extra education, t is year, r is the discount rate, T is expected years of working life, which one of the following is the best conclusion that can be made from the comparison of income flows and incurred costs of education?

- The rate of return in education will be higher whenever the discount rate is lower, the direct or indirect costs are lower, or the benefits are higher.
- If there is a higher rates of return in education for developing countries than that in developed countries, that has little to do with the average income differential between those with more and less schooling in developing and developed countries
- If private rate of return to a lower level of education is higher many families make this investment even if they have no ability to borrow.
- Education costs (direct tuition or other expenditures, and indirect costs, income forgone because the student could not work while in school) have lower weights as they come earlier than earning flows that come later.

Question Description	
Theme	Basic Macroeconomics
Course Name	Development economics II
Chapter	Chapter 3: Human Capital: Education and Health in Economic Development

Solution

The analysis of investments in health and education is unified in the human capital approach. Human capital is the term economists often use for education, health, and other human capacities that can raise productivity when increased. An analogy is made to conventional investments in physical capital: After an initial investment is made, a stream of higher future income can be generated from both expansion of education and improvements in health. As a result, a rate of return can be deduced and compared with returns to other investments. This is done by estimating the present discounted value of the increased income stream made possible by these investments and then comparing it with their direct and indirect costs. Of course, health and education also contribute directly to well-being.



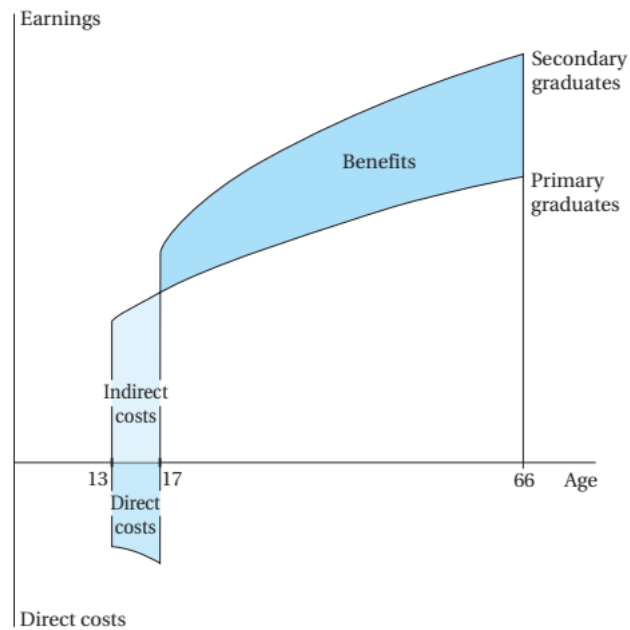


Figure showing the Financial Trade-offs in the Decision to continue in school

It is assumed that the individual works from the time he or she finishes school until he or she is unable to work, retires, or dies. This is taken to be 66 years. Two earnings profiles are presented—for workers with primary school but no secondary education and for those with a full secondary (but no higher) education. Primary graduates are assumed to begin work at age 13, secondary graduates at age 17. For an individual in a developing country deciding whether to go on from primary to secondary education, four years of income are forgone. This is the indirect cost, as labeled in the diagram. The child may work part time, a possibility ignored here for simplicity, but if so, only part of the indirect-cost area applies. There is also a direct cost, such as fees, school uniforms, books, and other expenditures that would not have been made if the individual had left school at the end of the primary grades. Over the rest of the person's life, he or she makes more money each year than would have been earned with only a primary education. This differential is labeled "Benefits" in the diagram. Before comparing costs with benefits, note that a dollar today is worth more to an individual than a dollar in the future, so those future income gains must be discounted accordingly, as is done in the Equation given in the question. The rate of return will be higher whenever the discount rate is lower, the direct or indirect costs are lower, or the benefits are higher.



Country Type and Region	Social Rate of Return (%)			Private Rate of Return (%)		
	Primary	Secondary	Higher	Primary	Secondary	Higher
Developing						
Sub-Saharan Africa	24	18	11	41	27	28
Asia	20	13	12	39	19	20
Latin America	18	13	12	26	17	20
Developed	14	10	9	22	12	12

This analysis was performed from the individual's point of view in the three right-hand columns of the table above. Notice that in sub-Saharan Africa and Asia, the private rate of return to primary education is about 40%! Despite this extraordinary return, many families do not make this investment because they have no ability to borrow even the meager amount of money that a working child can bring into the family. Note that the higher rates of return for developing countries reflect that the income differential between those with more and less schooling is greater on average than for the developed countries.

The first three columns of the table above indicate the social rate of return. This is found by including the amount of public subsidy for the individual's education as part of the direct costs, because this is part of the investment from the social point of view. It should be noted that these social returns are probably understated because they do not take into account the externality that educated people confer on others (e.g., being able to read for other family members), not to mention other individual and social benefits such as increased autonomy and civic participation.

Hence the correct answer is **(A)** The rate of return in education will be higher whenever the discount rate is lower, the direct or indirect costs are lower, or the benefits are higher.

